

PHD PROGRAM

Applied Data Science & Artificial Intelligence



UNIVERSITÀ
DEGLI STUDI
DI TRIESTE

Explainable Artificial Intelligence

XAI: AN OVERVIEW



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APPLIED DATA SCIENCE &
ARTIFICIAL INTELLIGENCE

The concept of explainability

XAI aims at explaining model predictions.

Explanations should carry information for humans.

Models should be interpreted by humans.

Models predictions should be understandable.

What is an explanation?

What kind of information?
For which purpose?

How a model can be interpreted?
By whom?

What means to understand a prediction?
Who has to understand it?

Z.C. Lipton. The mythos of model interpretability, 2018.

Model interpretability: objective or subjective?

Model interpretation defined by its **usefulness**.

Useful in which sense?

- It provides useful information.
- It uncovers causal relationship in data.
- It generates scientific understanding.
- It is informative for decision makers.

Model interpretation as the **degree of trust** in a model.

Trust in which sense?

- It is accurate.
- It makes human-like mistakes: we can give control to the model.
- It gives us confidence that the model is appropriate: it works for the *real* task.
- It is robust to changing or adversarial environments.
- It is fair / unbiased.

A formal definition?

There is no agreed-upon definition of what is an explanation or interpretation for a model.

Is the very notion of explainable artificial intelligence ill-posed?

The concept of explainability

Interpretability

The ability to understand the decision-making process of the model

Complete understanding of the relationship between input and output

Key word: transparency

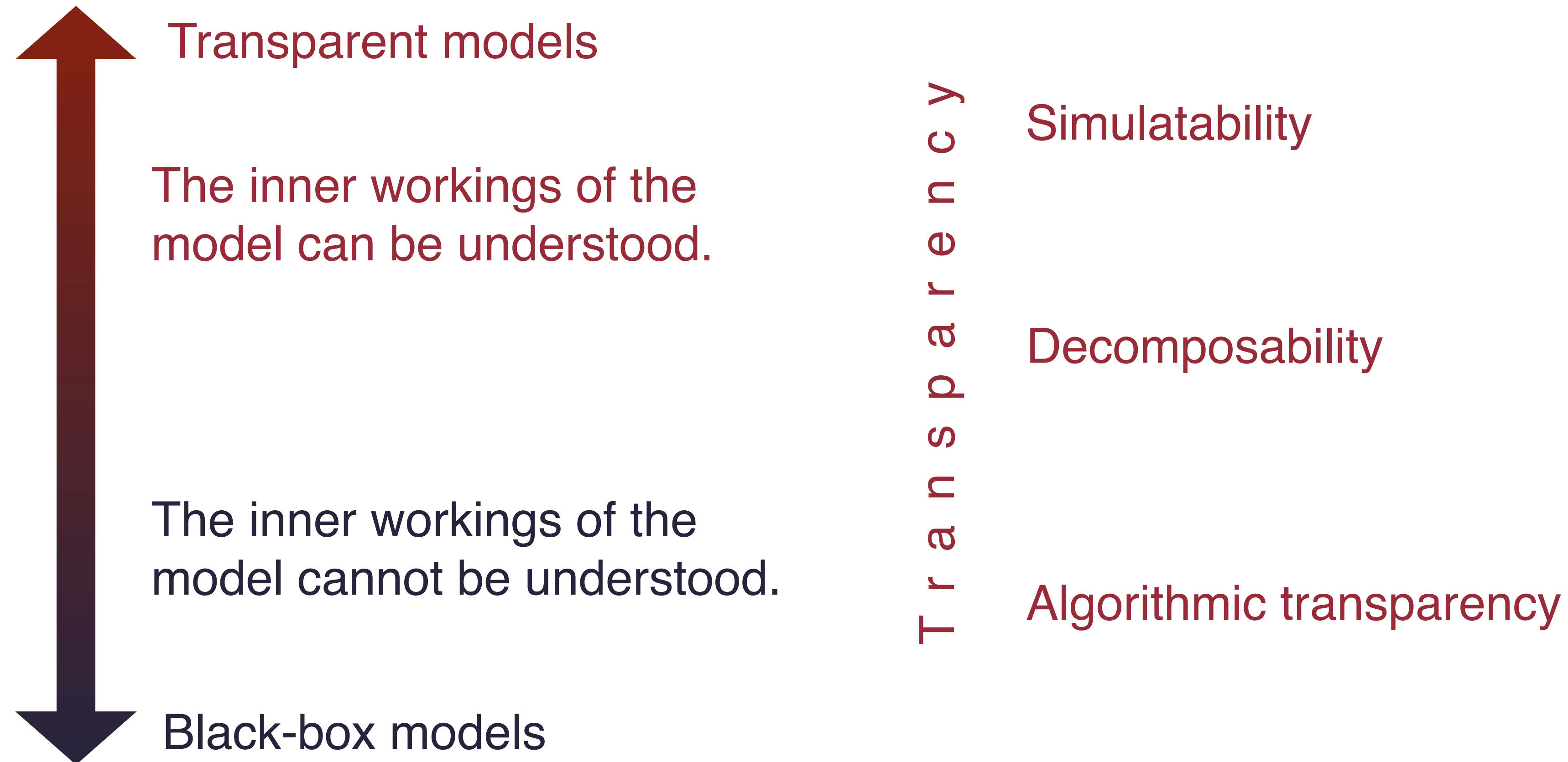
Explainability

The ability of explain the decision-making process of the model in a simple and understandable way to the user

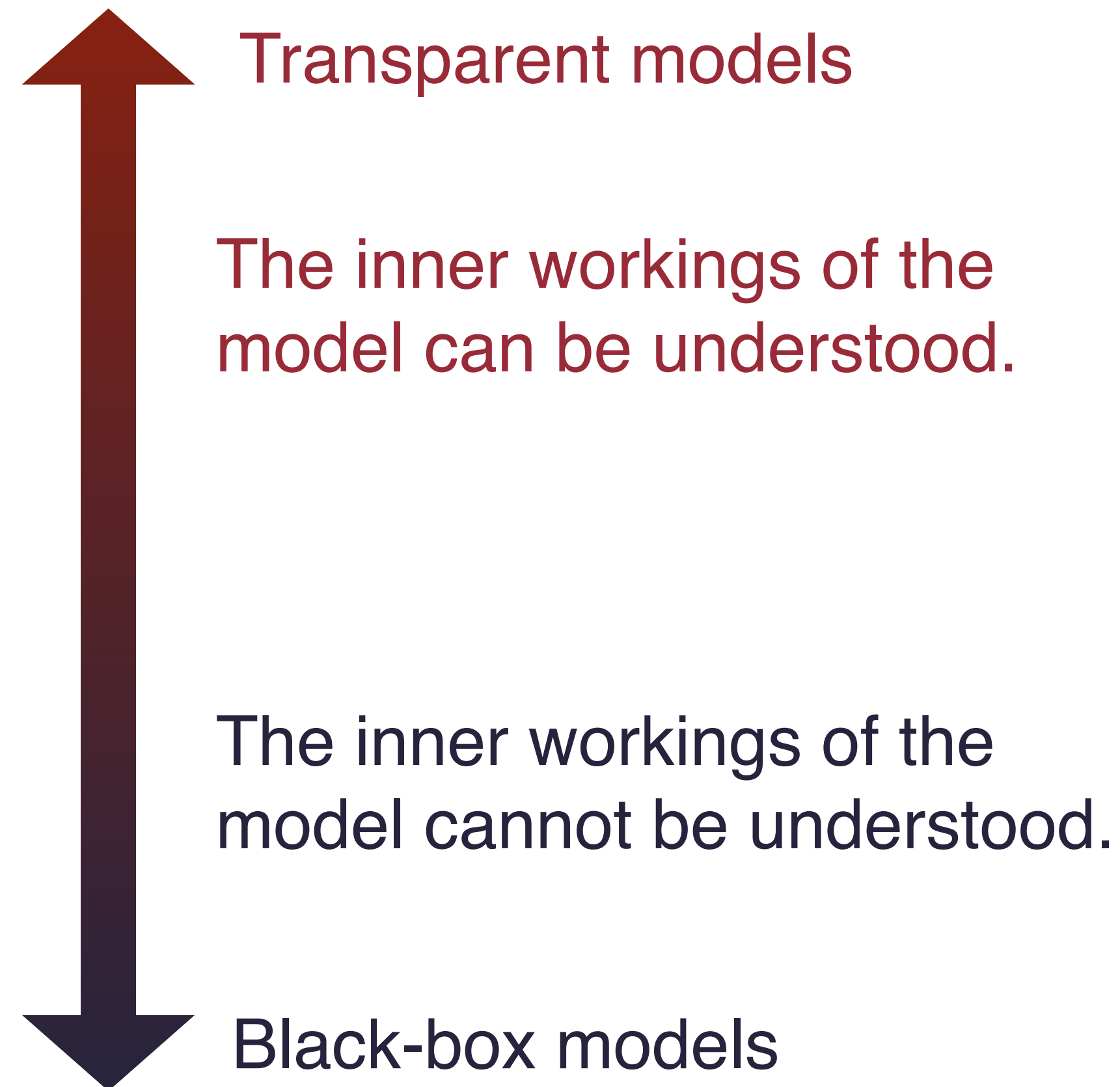
Intuition of the relationship between input and output

Key word: comprehension

Transparency vs Black-Box



Transparency vs Black-Box



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Simulatability

A model that can be contemplated and/ or simulated by a human: linear regression vs xgboost.

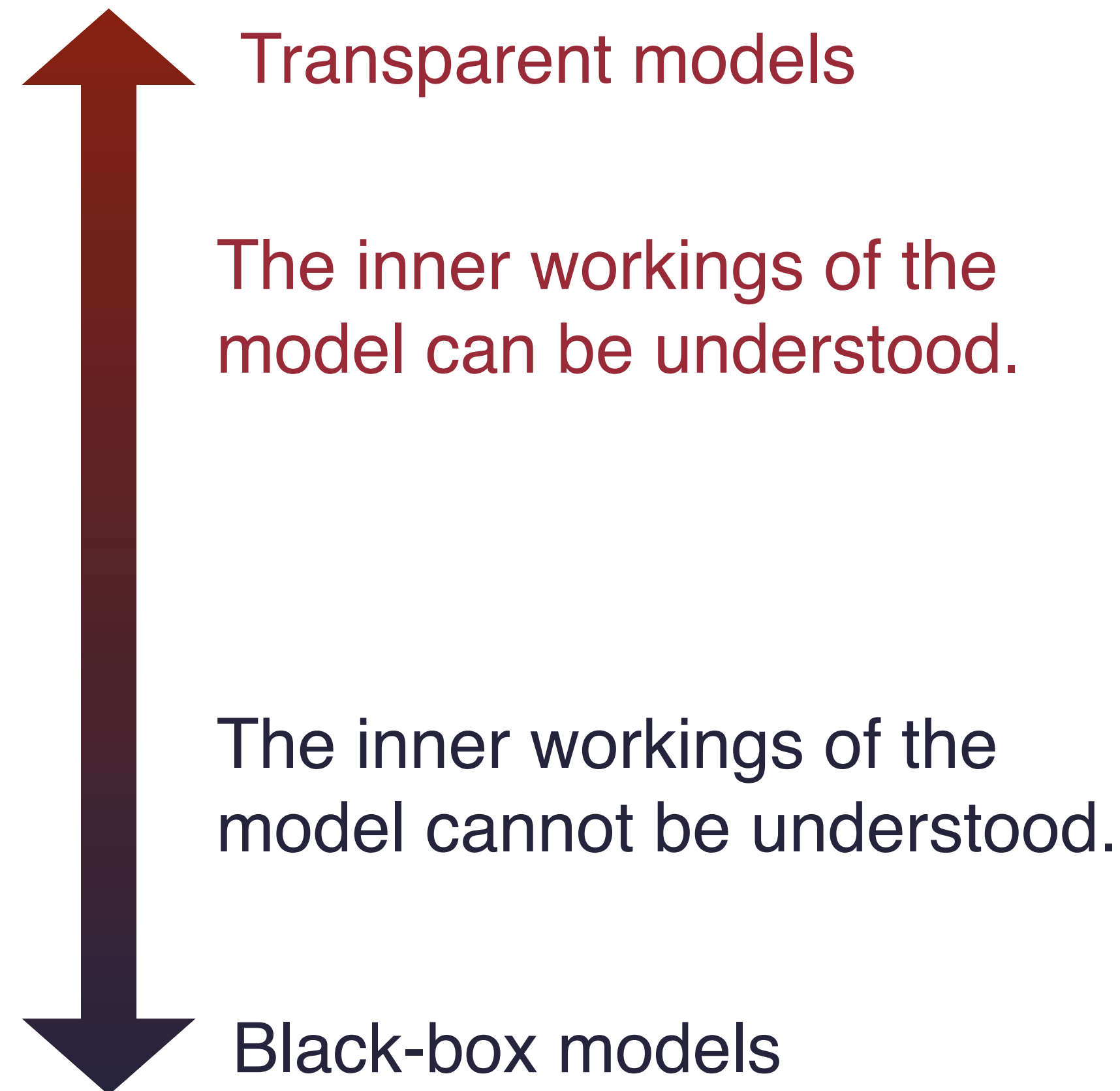
For understanding: model size need to be small. Linear regression with 1000 variables is not simulatable.

Model inference needs to be (computationally) simple: moderate sizer decision trees may work.

Decomposability

Algorithmic transparency

Transparency vs Black-Box



T r a n s p a r e n c y

Simulatability

Decomposability

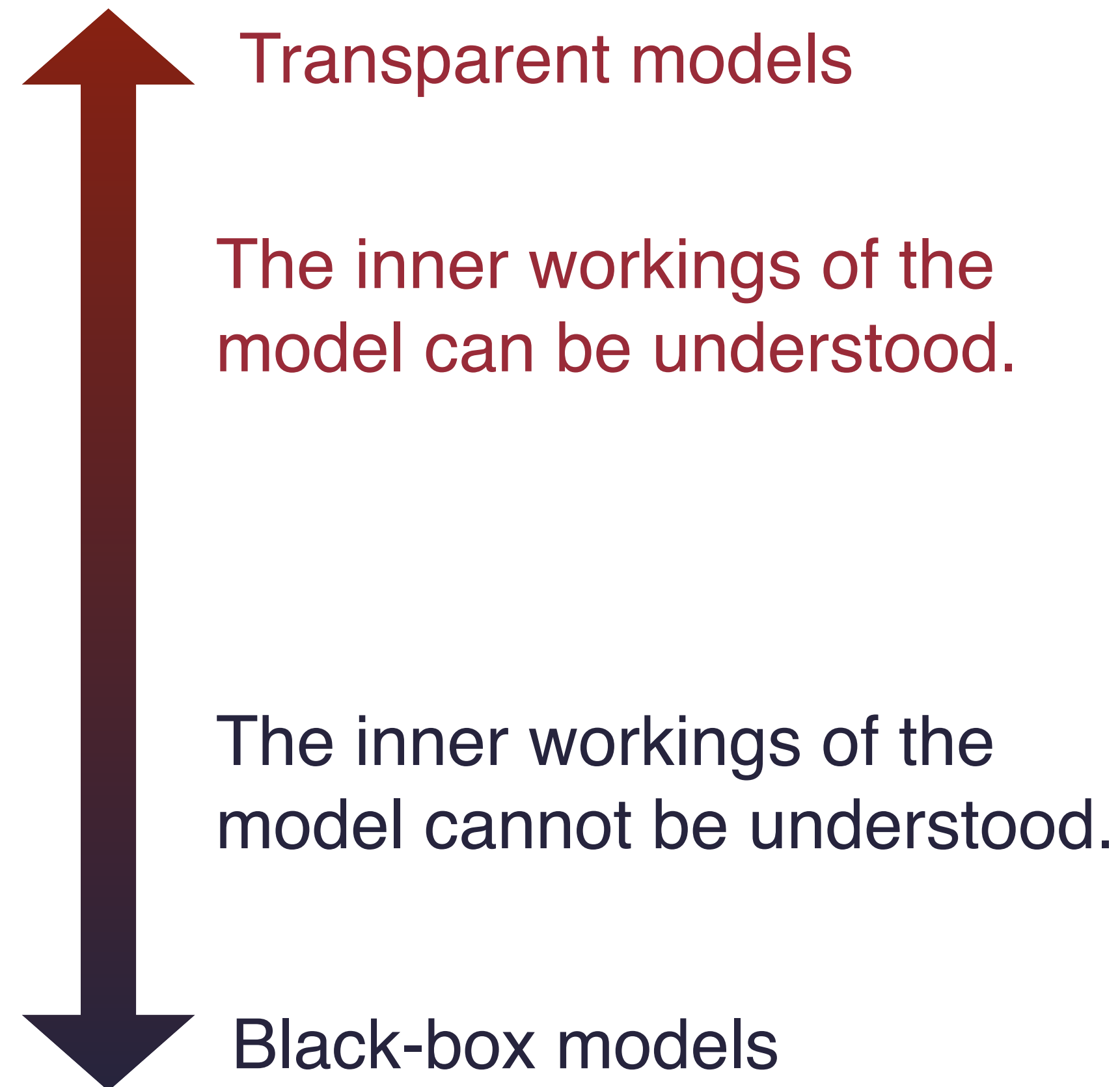
Each component of a model (input, parameters, calculations) admits an intuitive explanation.

Linear regression and decision trees will do.
Limit: no feature pre-processing

Beware of robustness w.r.t. model components (e.g. feature selection)

Algorithmic transparency

Transparency vs Black-Box



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Simulatability

Decomposability

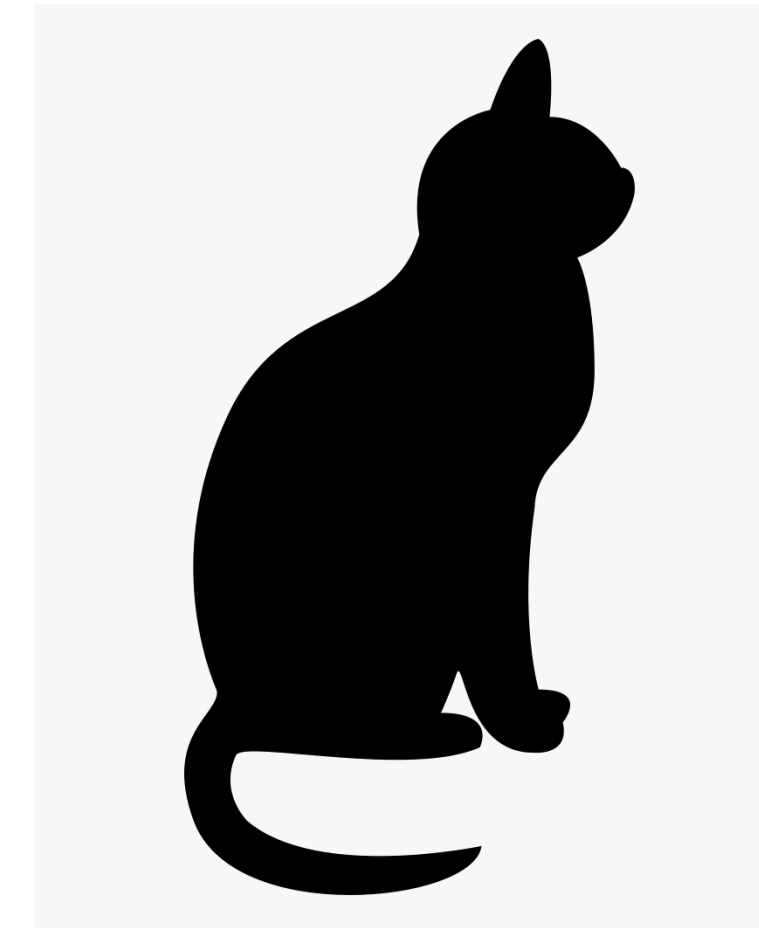
Algorithmic transparency

The learning algorithms has an understandable mechanics.

E.g. the optimization for linear regression is simple to understand (convex loss function), while that for deep neural networks is not.

Human explanations

What do you see?



Functioning of human mind is not transparent at all. Our explanations (of cognitive tasks) are just build after the task has been performed.

We will call such an approach **post-hoc explanation**.

Post-hoc interpretability

It is a technique that seeks to extract information from a learned model, mimicking human-like explanations

You do not explain how a model worked, but try to justify model decisions.

There are several different approaches.

Numeric explanations

Textual explanations

Visual explanations

Rules-based explanations

Mixed explanations

G. Vilone and L. Longo, Classification of Explainable Artificial Intelligence Methods through Their Output Formats, 2021

Post-hoc interpretability

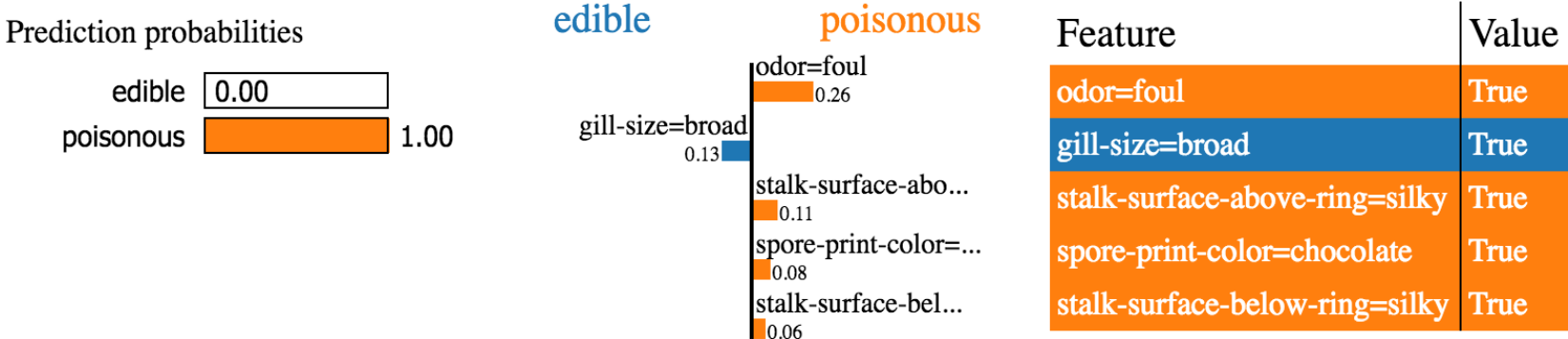
It is a technique that seeks to extract information from a learned model, mimicking human-like explanations

You do not explain how a model worked, but try to justify model decisions.

There are several different approaches.

Numeric explanations

Generate crisp values, vectors of numbers, matrices, or tensors to highlight the input attributes or features of a model that have the largest effect on the prediction of the output.



- Textual explanations
- Visual explanations
- Rules-based explanations
- Mixed explanations

Ribeiro et al. Model-Agnostic Interpretability of Machine Learning , 2017.

Post-hoc interpretability

It is a technique that seeks to extract information from a learned model, mimicking human-like explanations

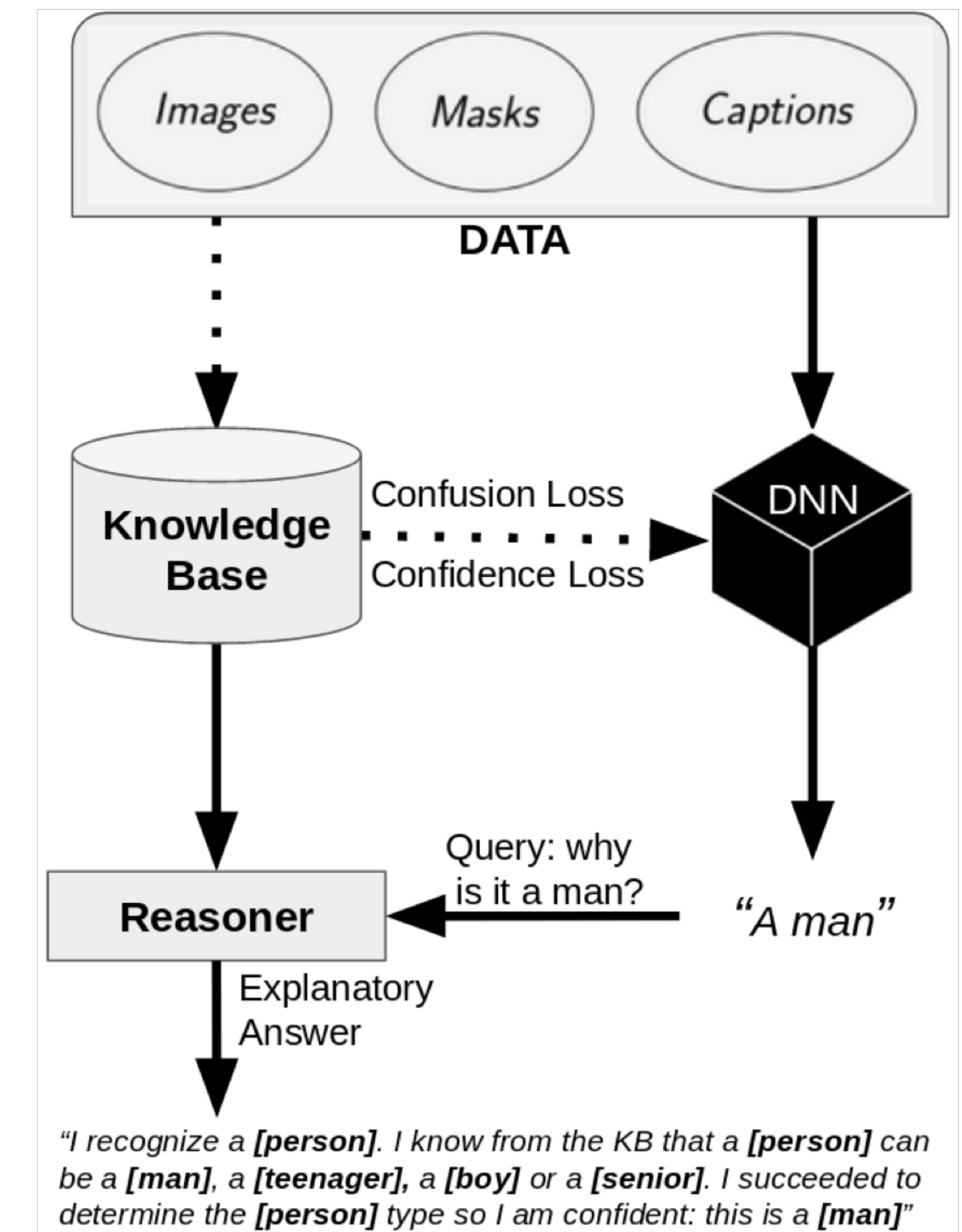
You do not explain how a model worked, but try to justify model decisions.

There are several different approaches.

Numeric explanations Textual explanations

Generate natural language statements that can either be written or orally uttered, providing a human-readable and intuitive format for explanations.

Visual explanations Rules-based explanations Mixed explanations



A. Bennetot et al. Towards Explainable Neural-Symbolic Visual Reasoning, 2019.

Post-hoc interpretability

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You do not explain how a model worked, but try to justify model decisions.

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Salvaraju et al. Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization, 2020.

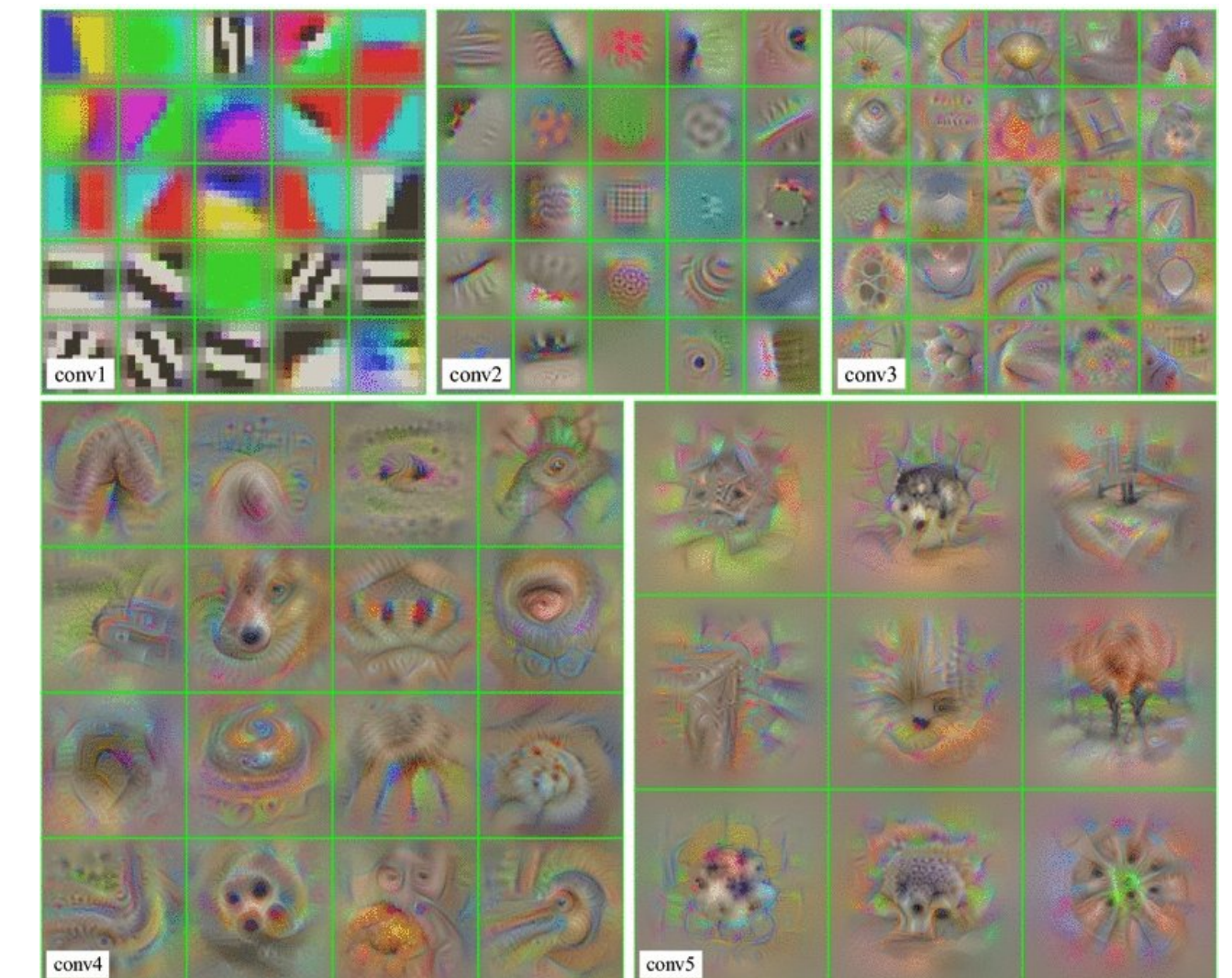
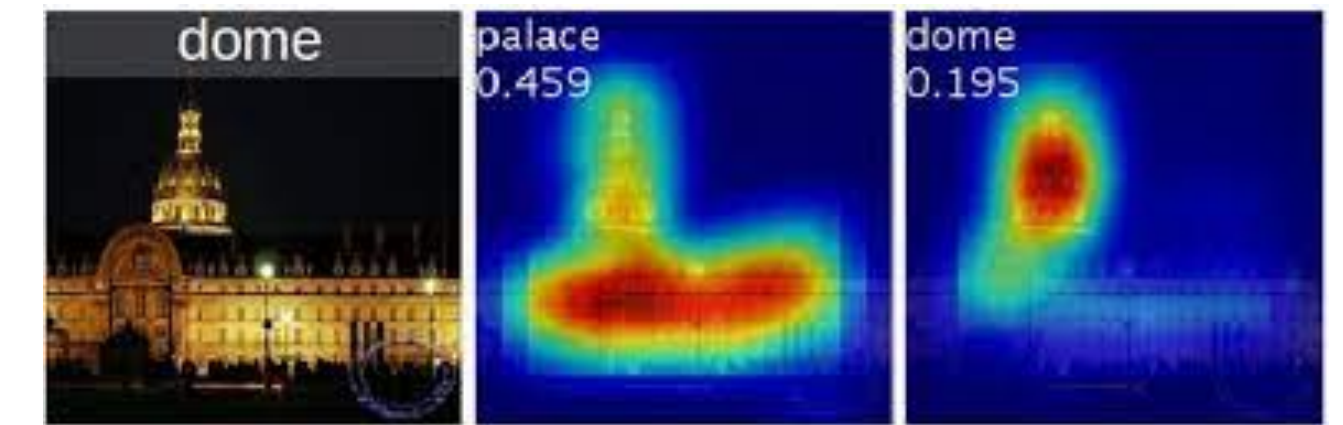
A. Nguyen et al. Understanding Neural Networks via Feature Visualization: A survey, 2019.

Numeric explanations

Textual explanations

Visual explanations

Use graphical tools to illustrate information, often through heat-maps, graphs, or other visualizations that highlight specific areas of an image or text that influence the model's inferential process.



Rules-based explanations

Mixed explanations

Post-hoc interpretability

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Numeric explanations

Textual explanations

Visual explanations

Rules-based explanations

Use a schematic, logical format, typically in the form of “IF...THEN” statements with AND/OR operators, to express combinations of input features and their activation values.

IF Country = United-States
AND Capital Loss = Low
AND Race = White **AND**
Relationship = Husband **AND**
Married **AND** $28 < \text{Age} \leq 37$
AND Sex = Male **AND** High
School grad **AND** Occupation
= Blue-Collar **THEN**
PREDICT Salary > \$50K

Mixed explanations

M.T. Ribeiro et al. Anchors: High-precision model-agnostic explanations, 2018.

Post-hoc interpretability

It is a technique that seeks to extract information from a learned model, mimicking human-like explanations

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There are several different approaches.

Numeric explanations

Textual explanations

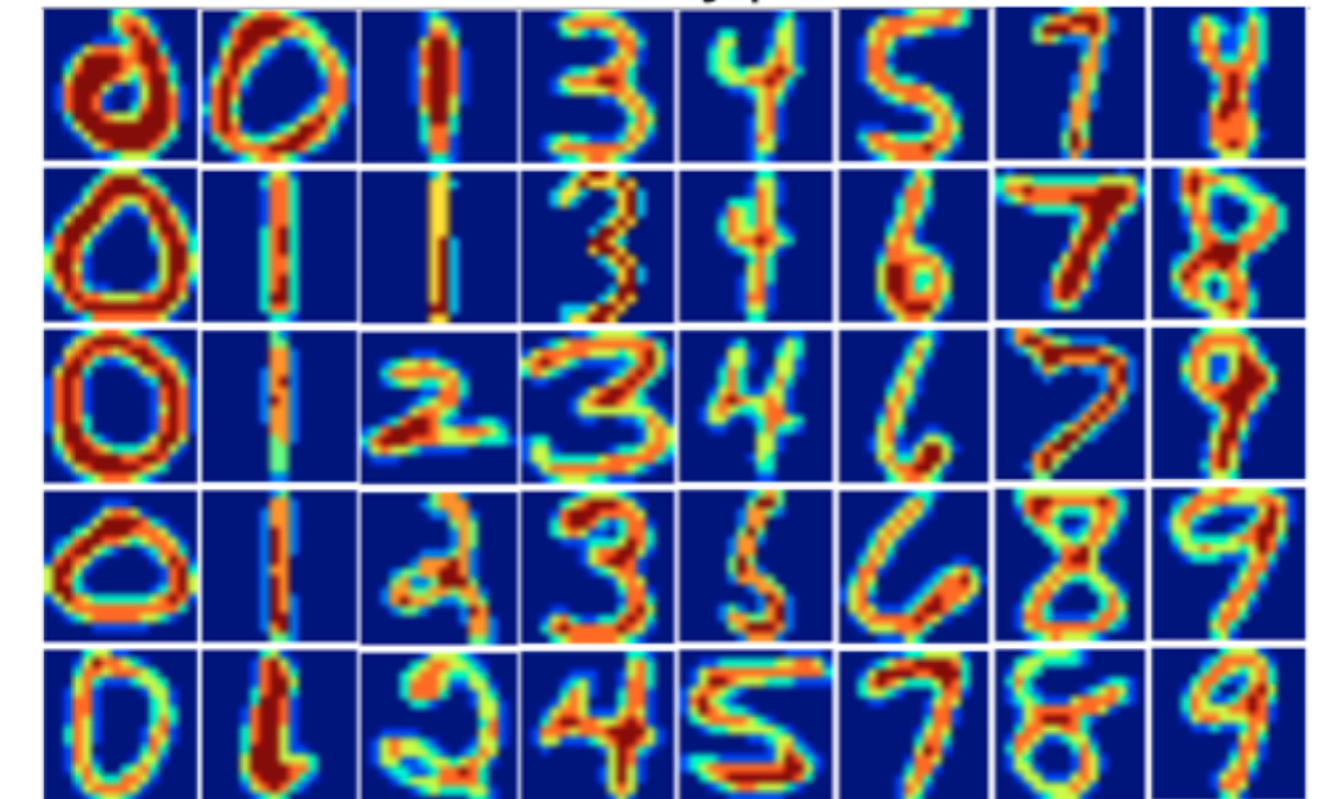
Visual explanations

Rules-based explanations

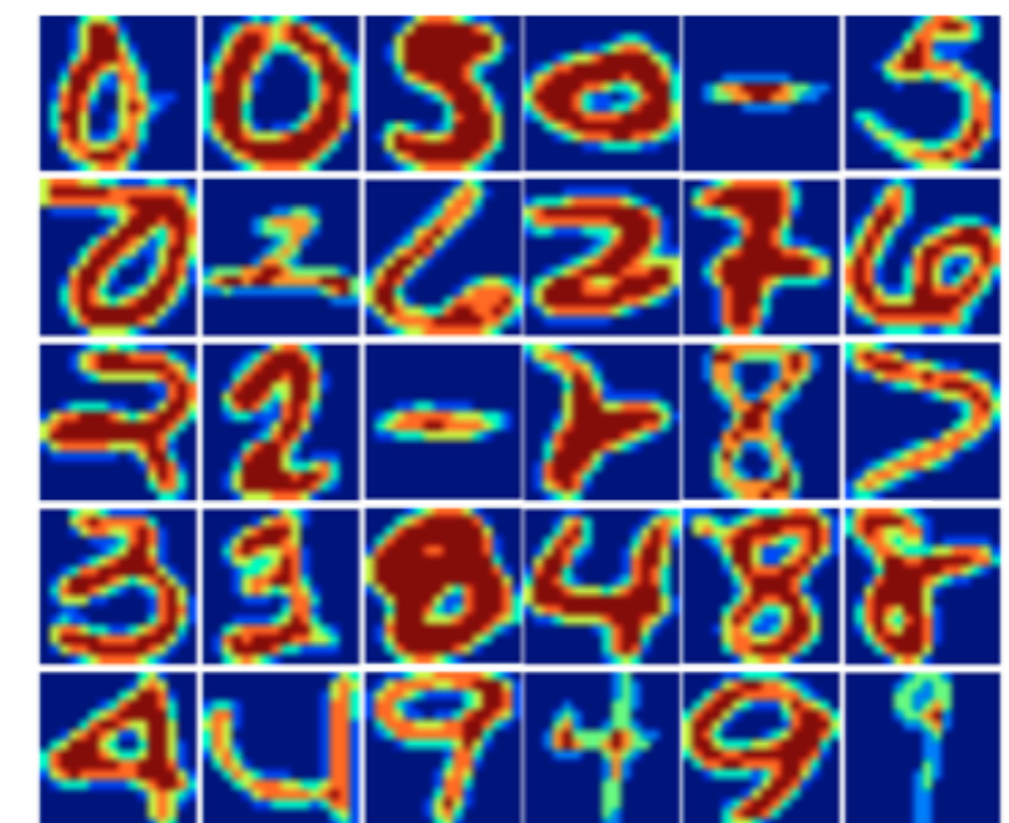
Mixed explanations

Combine multiple formats, such as visual, textual, and numeric explanations, to exploit their strengths and overcome individual weaknesses.

Prototypes



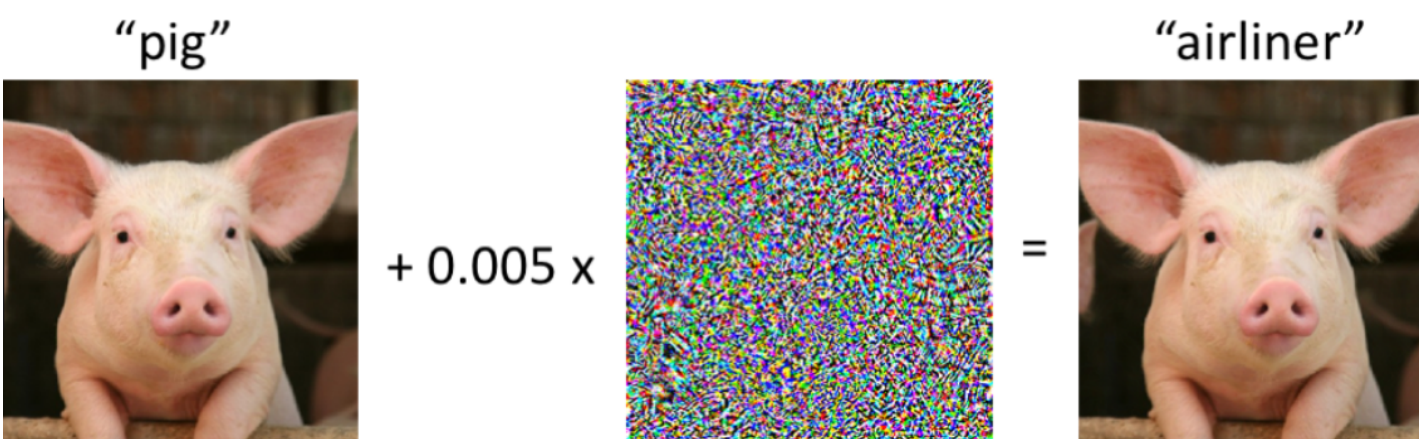
Criticisms



B. Kim et al. Examples are not enough, learn to criticize! criticism for interpretability, 2016

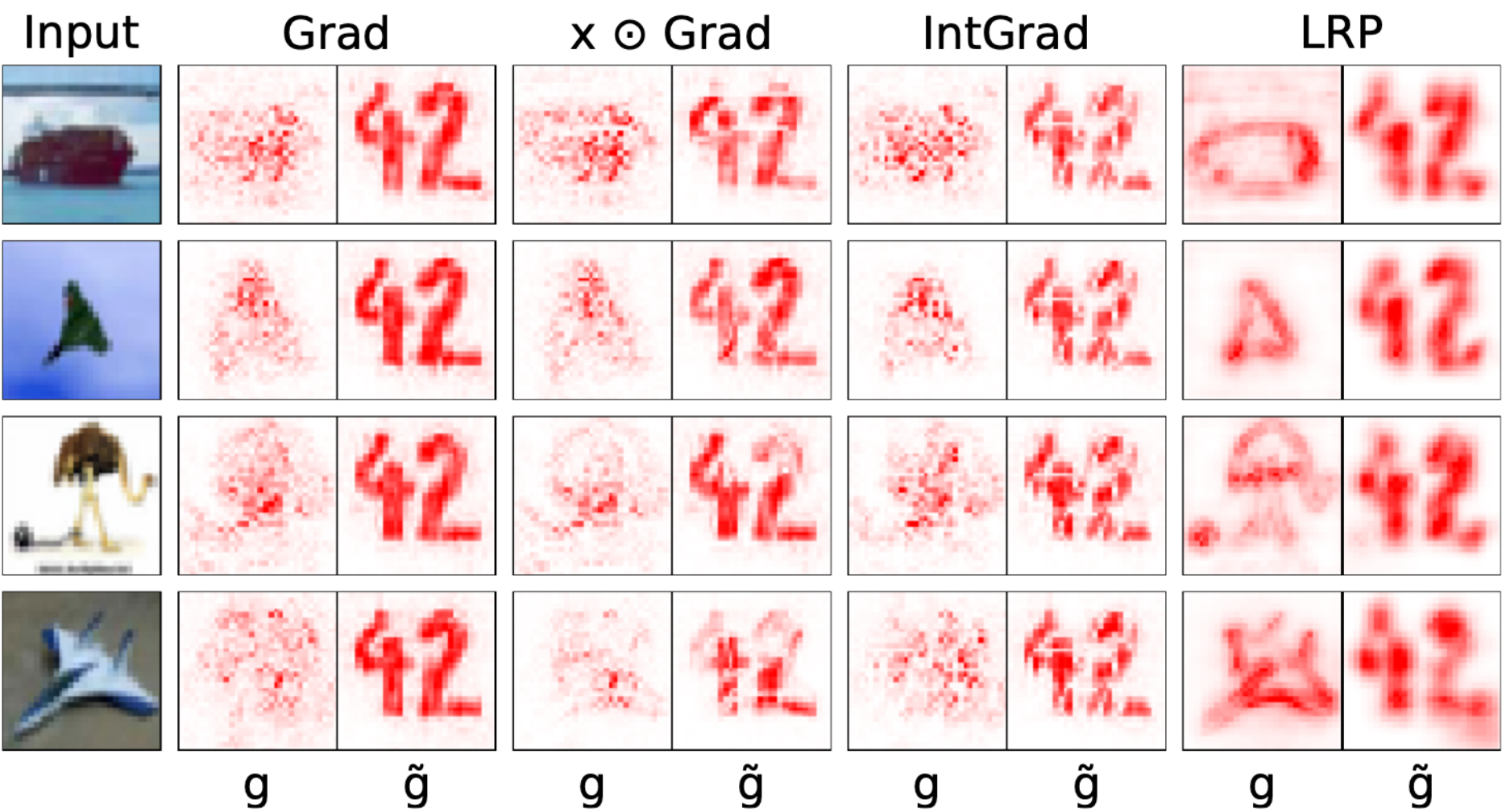
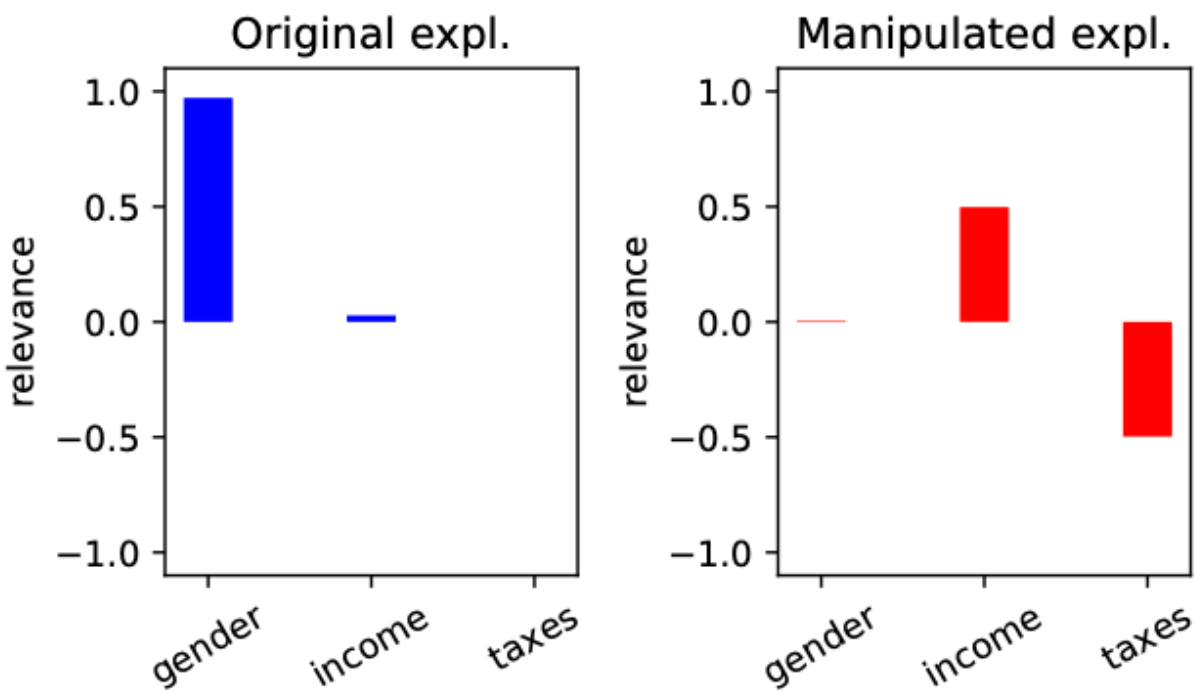
Trusting post-hoc explanations

Robustness of models in making predictions is a source of concern.



But are post-hoc explanations robust?

Neither to random perturbations nor (more worryingly) to adversarial attacks...



Anders et al. Fairwashing Explanations with Off-Manifold Detergent, 2020.

The limits of simplicity

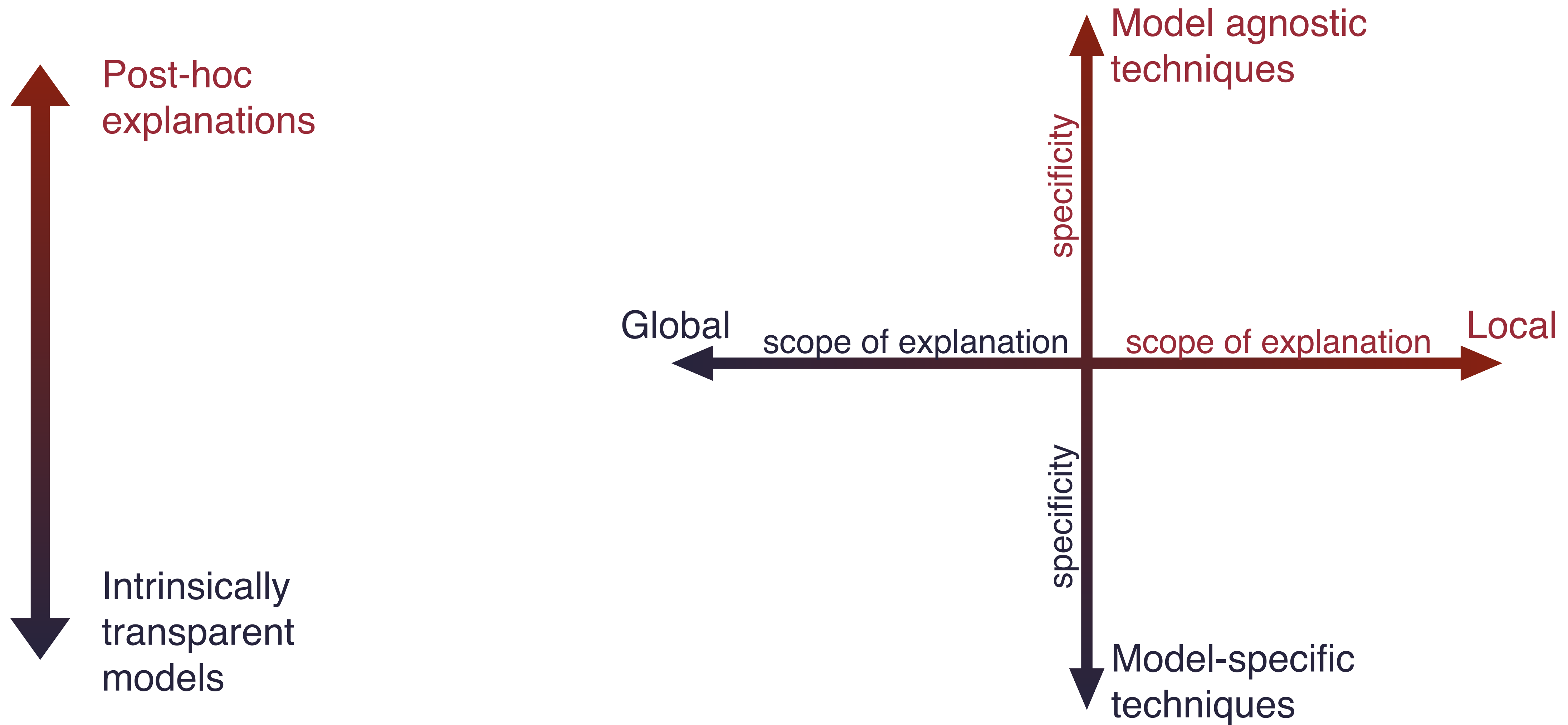
Transparency (simulatability) is intrinsically requiring models to be **simple**, due to intrinsic limits of us as humans.

This rises a **trade-off** between interpretability and model performance

Moreover, simple models are essentially just very few: most models are complex. Hence only few problems can admit a transparent model which is also accurate.

In this light, this concept of transparency is not suitable in practice. This means that simple logic rules are not effective unless problems are simple.

A taxonomy of XAI



A taxonomy of XAI methods

global, model-agnostic

local, model-agnostic

LIME

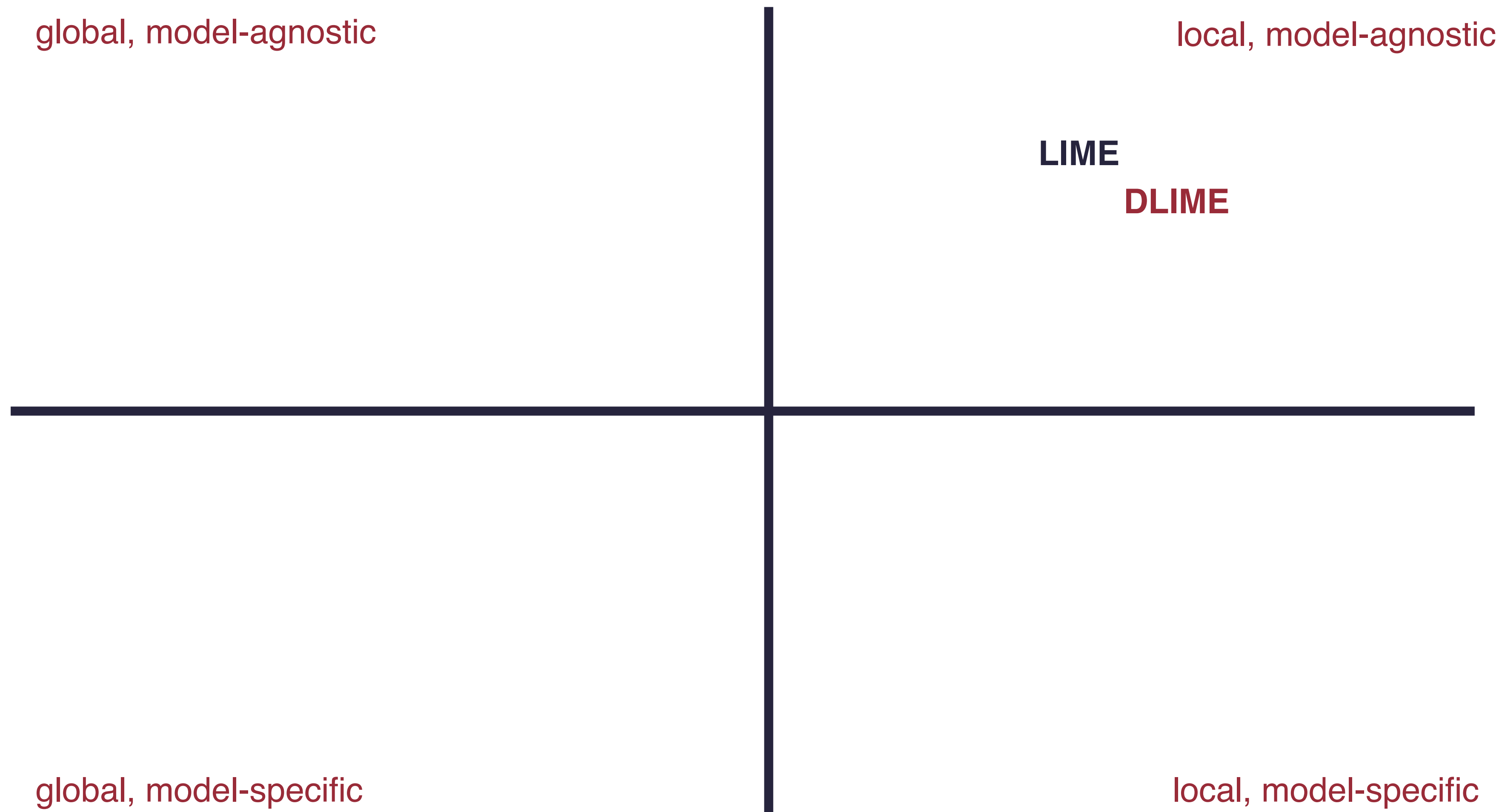
Local Interpretable Model-agnostic Explanations.

Locally fitting an interpretable model (sparse linear regression, decision tree) on a random set of points sampled around the point of interest. Featuring kernel weighting and feature transformation.

global, model-specific

local, model-specific

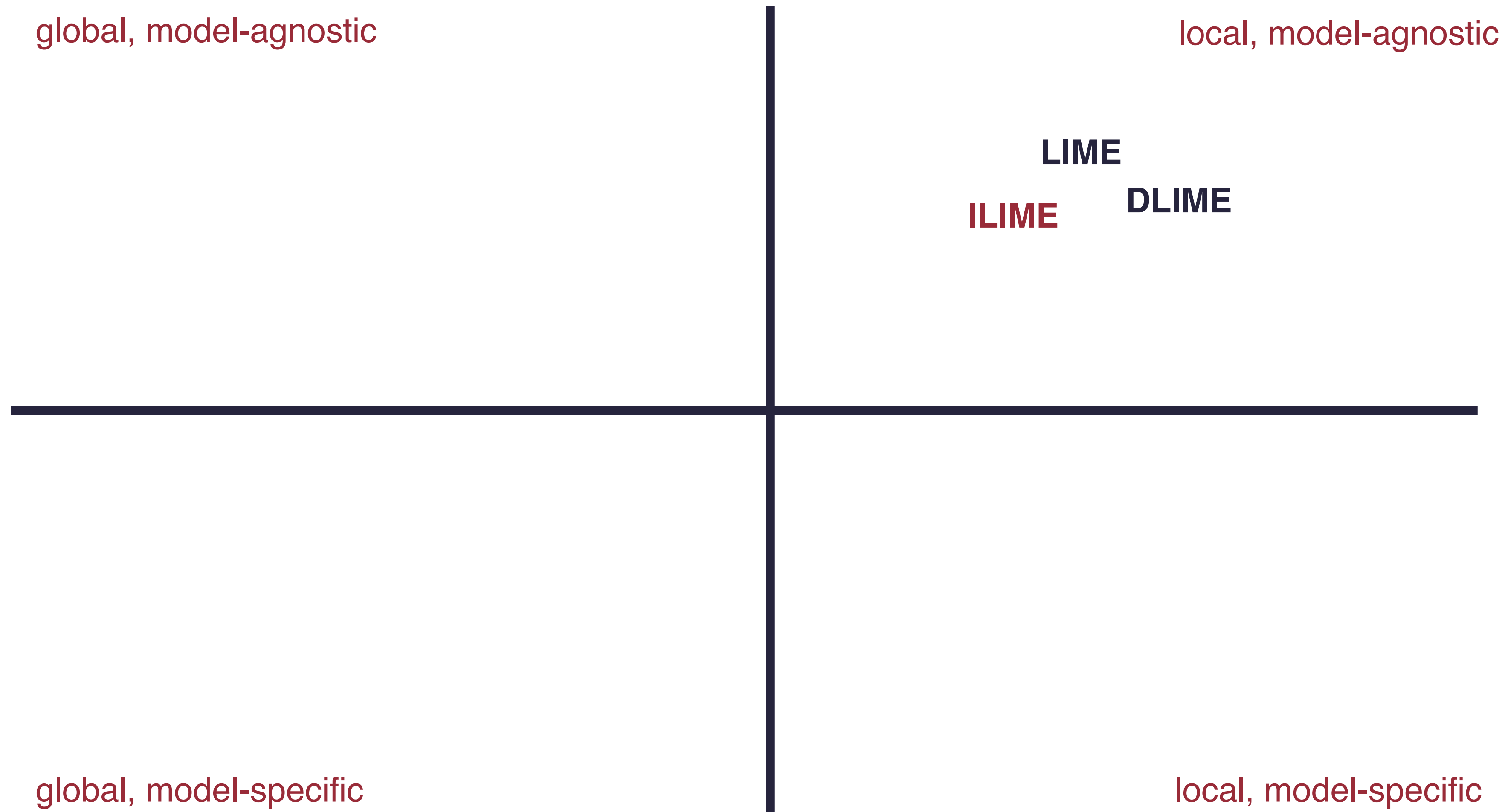
A taxonomy of XAI methods



Deterministic LIME.

Overcome sensitivity to random choice of points by hierarchically clustering the dataset and using LIME on the closest cluster to a new point (by KNN).

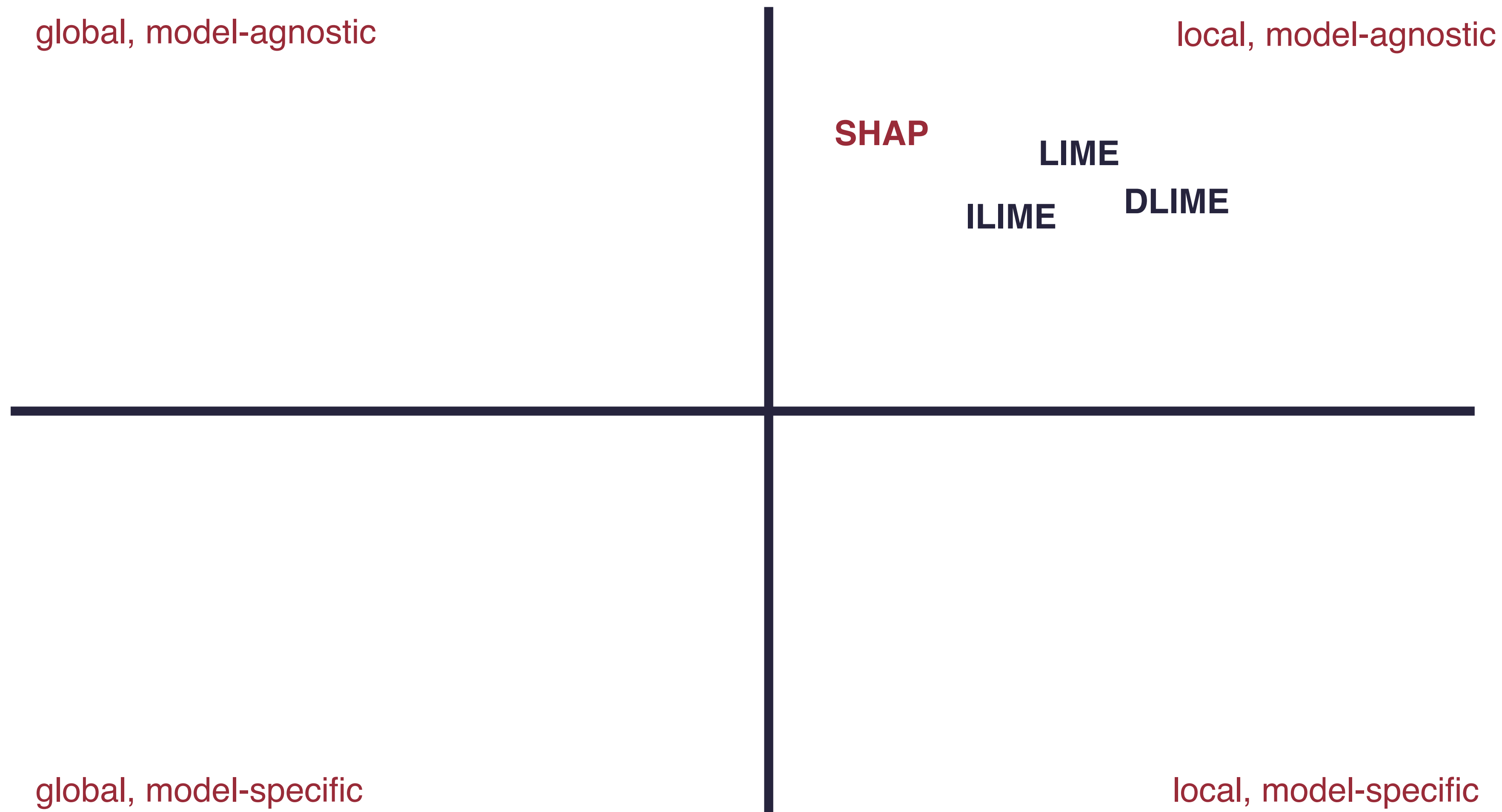
A taxonomy of XAI methods



ILIME: LIME with influences.

Uses a notion of influence to refine the weighting of points, to reduce sensitivity to choice of the random dataset. This biases the linear model that is learned. Influences are proportional to the gradient loss.

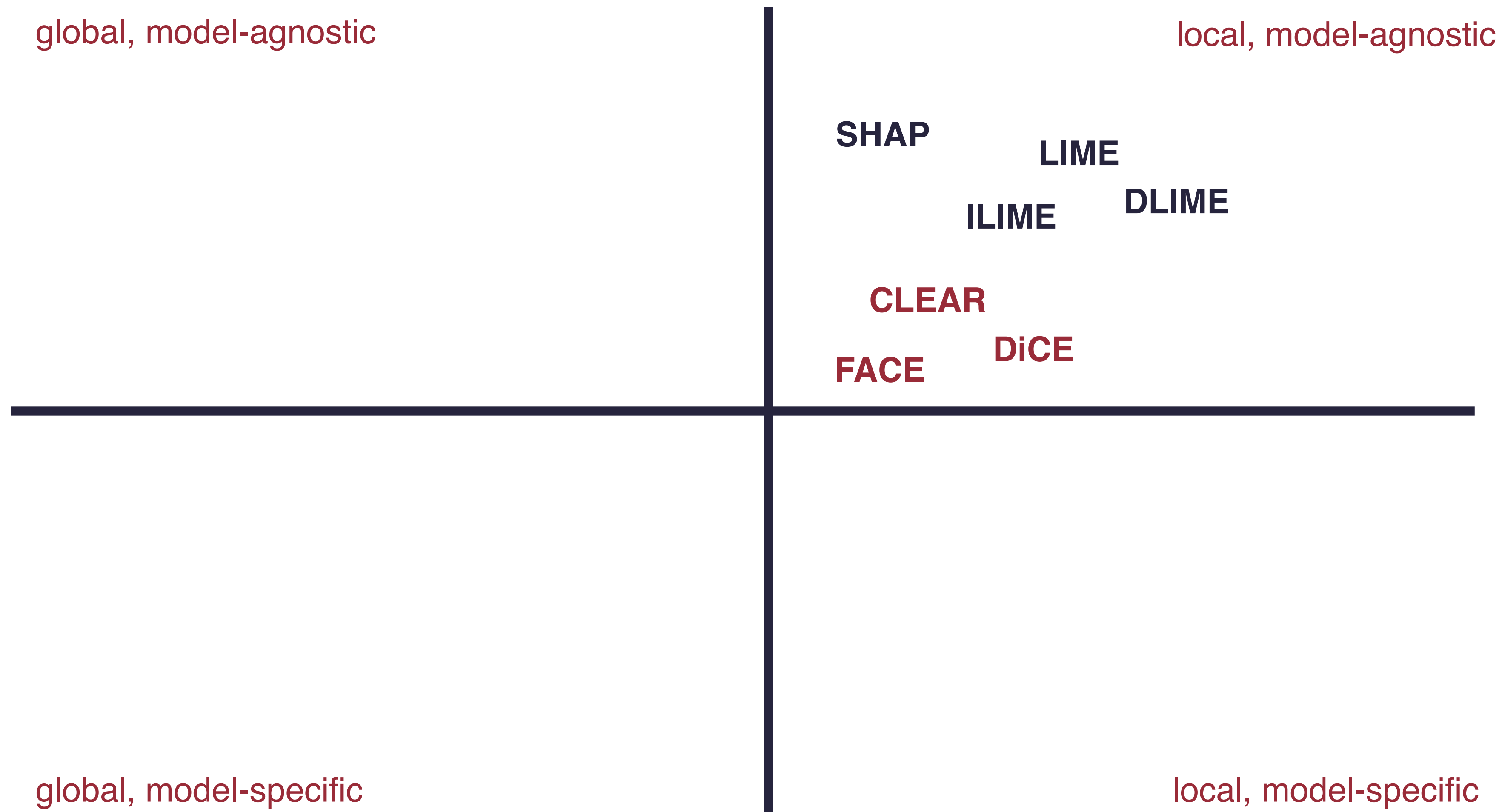
A taxonomy of XAI methods



SHapley Additive
exPlanation.

SHAP values describe the contribution of each feature in the prediction by considering what is the effect of adding a feature to any subset of features w.r.t. model prediction.

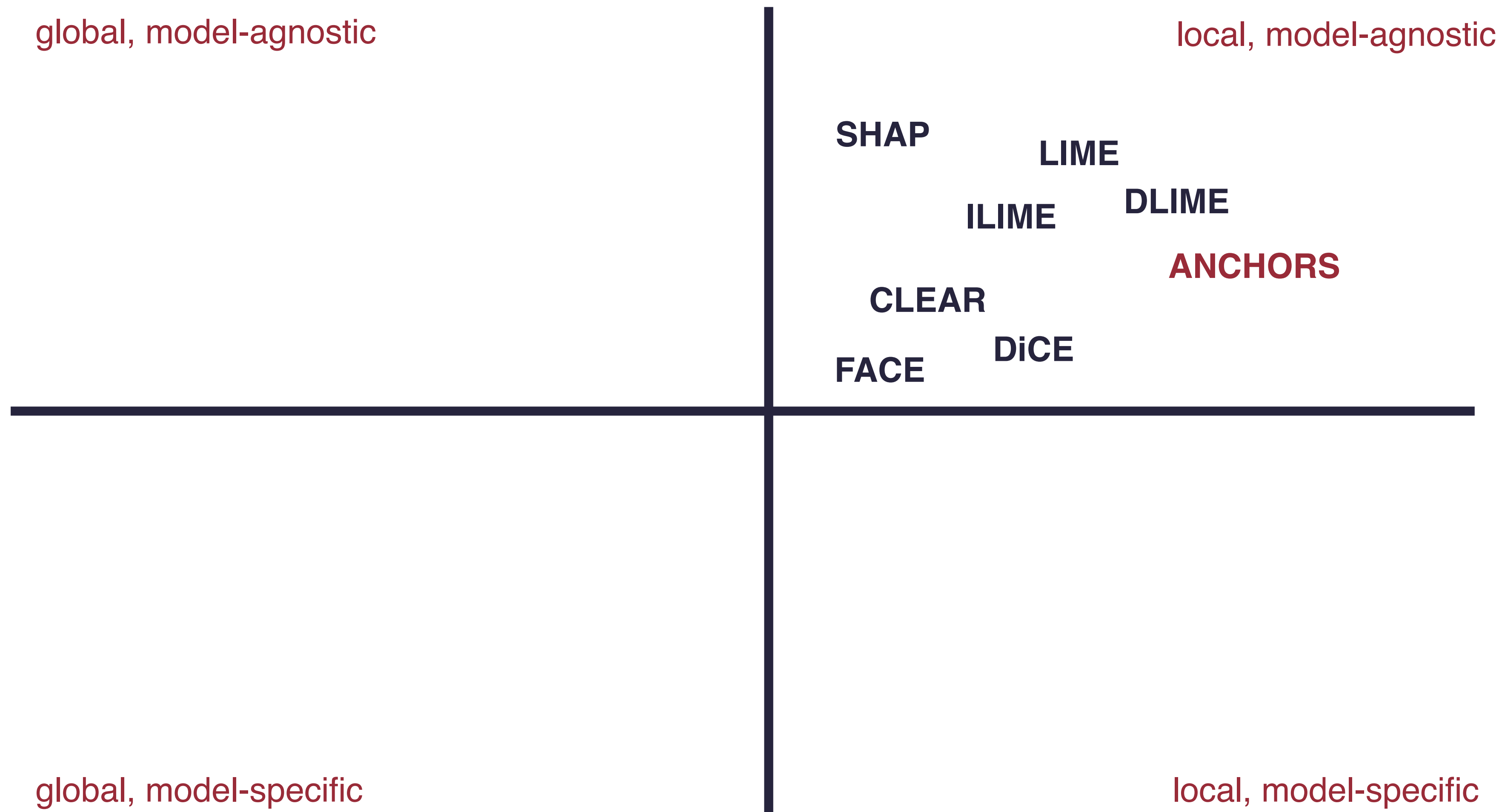
A taxonomy of XAI methods



Counterfactual Local Explanations; Diverse Counterfactual Explanations; Feasible and Actionable Counterfactual Explanations.

These are methods to generate counterfactual explanations, showing a set of minimally perturbed examples with possible constraints on feasibility, proximity, diversity.

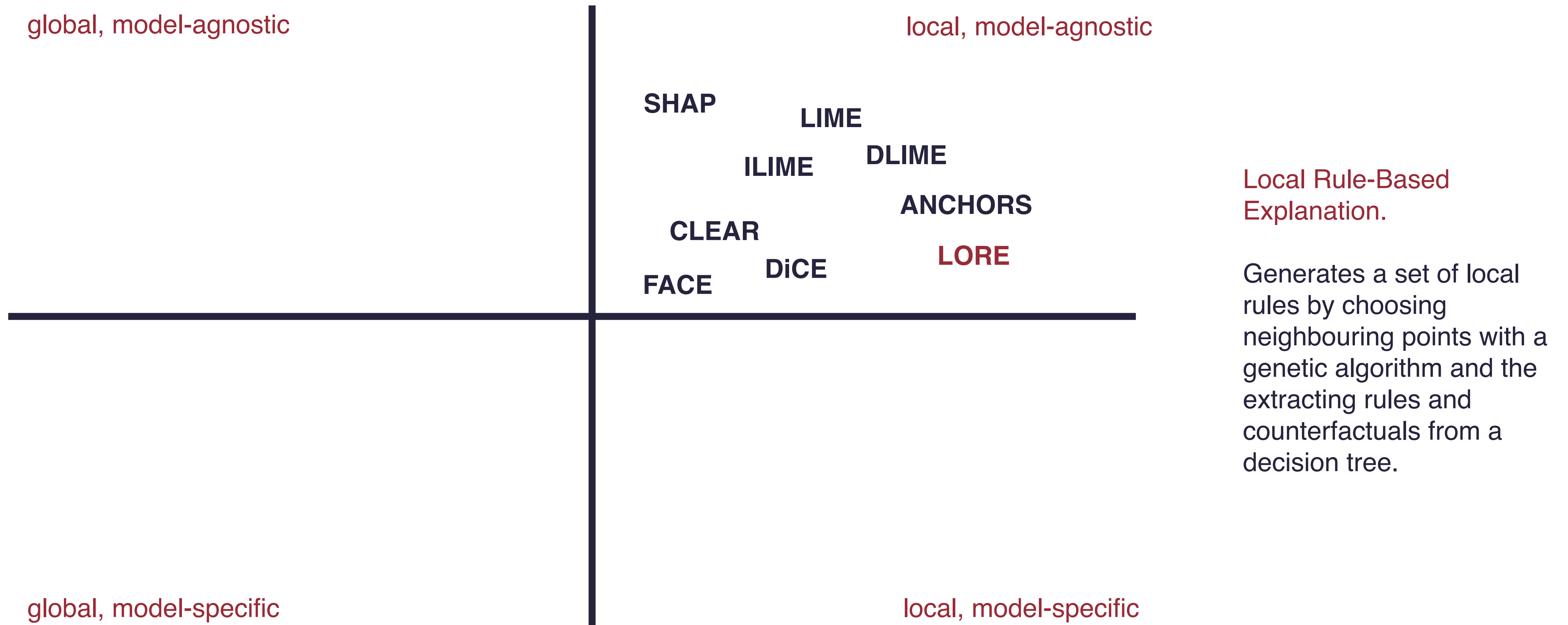
A taxonomy of XAI methods



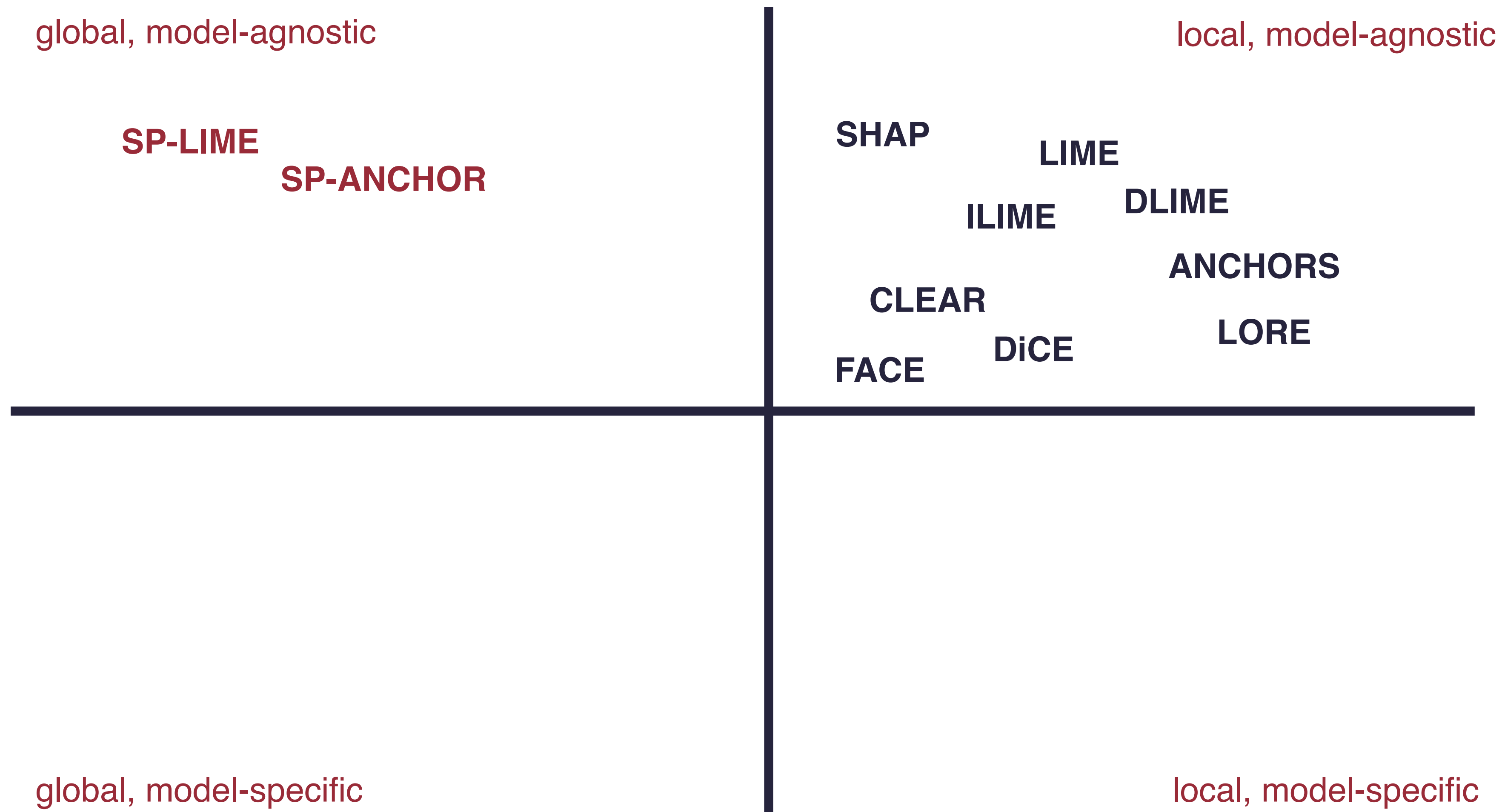
ANCHORS.

It's a local approximation constructing a set of rules that approximates locally around x the predictor.

A taxonomy of XAI methods



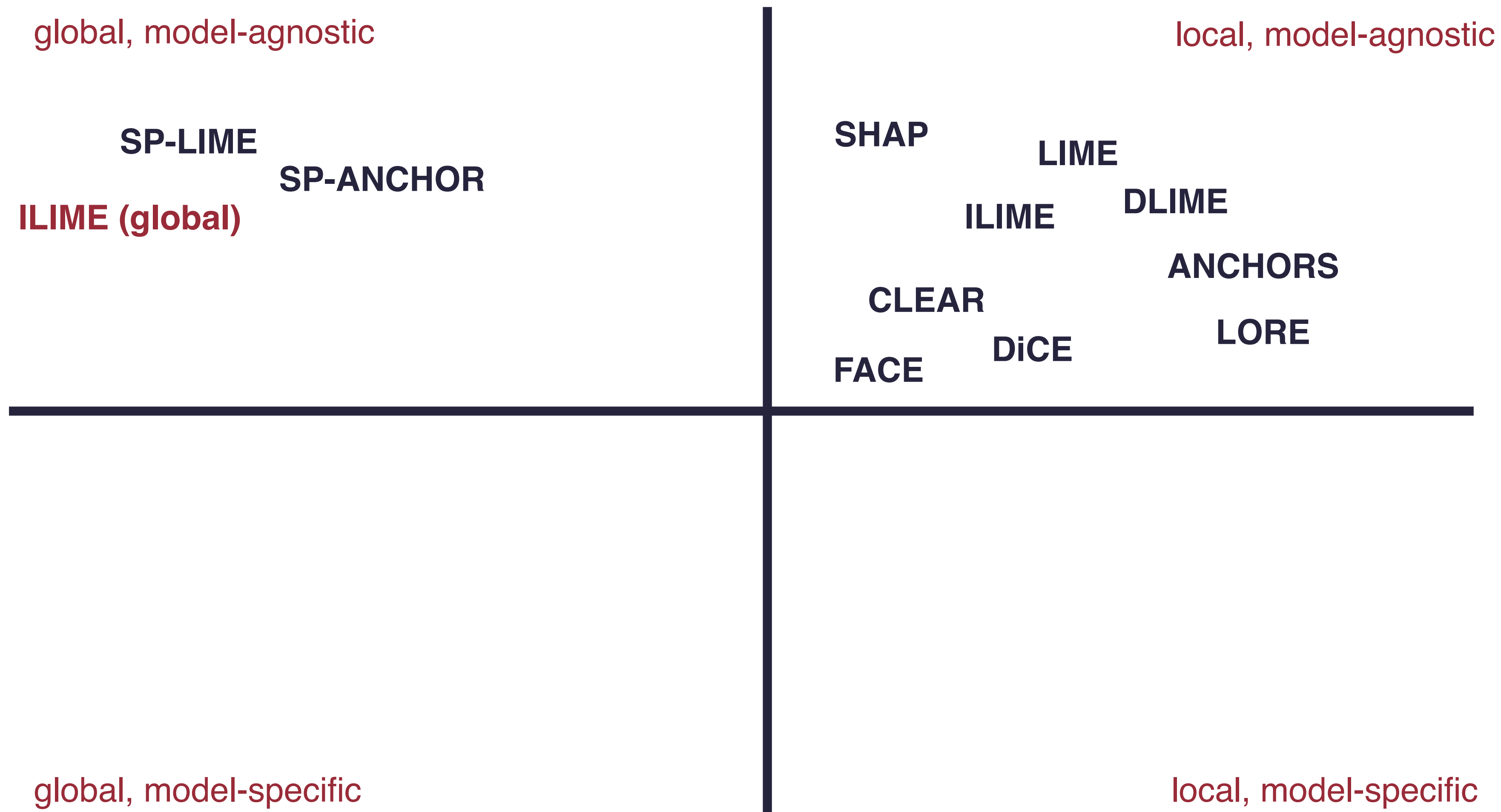
A taxonomy of XAI methods



Submodular Pick LIME;
Submodular Pick Anchor

Extends LIME and Anchors to global explainability by identifying a set of instances whose combined local explanations cover most of the model behaviour. They use a submodular strategy to pick the right points.

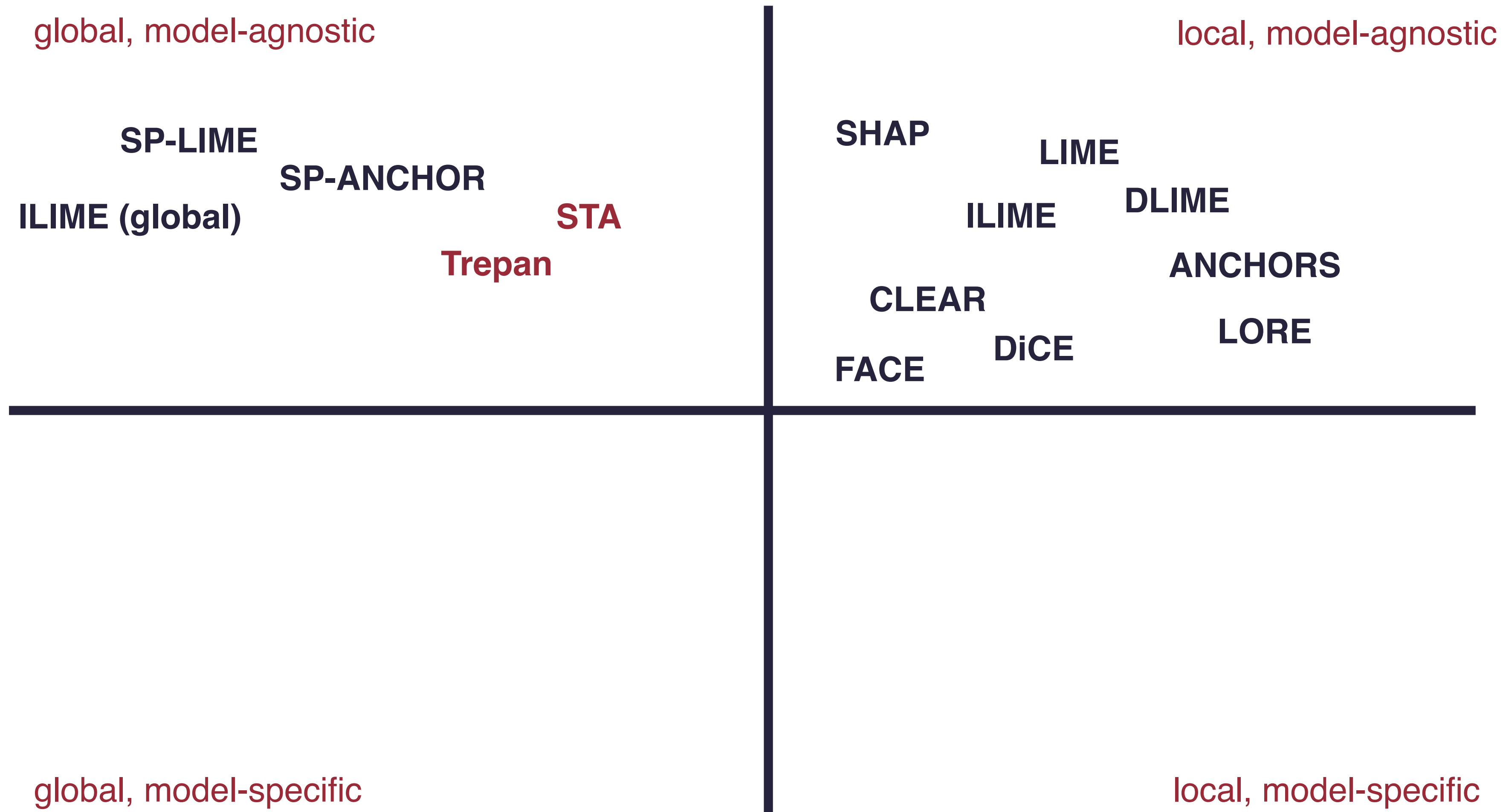
A taxonomy of XAI methods



Global ILIME

Extension of ILIME using a clustering algorithm to identify clusters in local model explanations (one for each point in the dataset), in order to identify the best number of examples to cover the decision space.

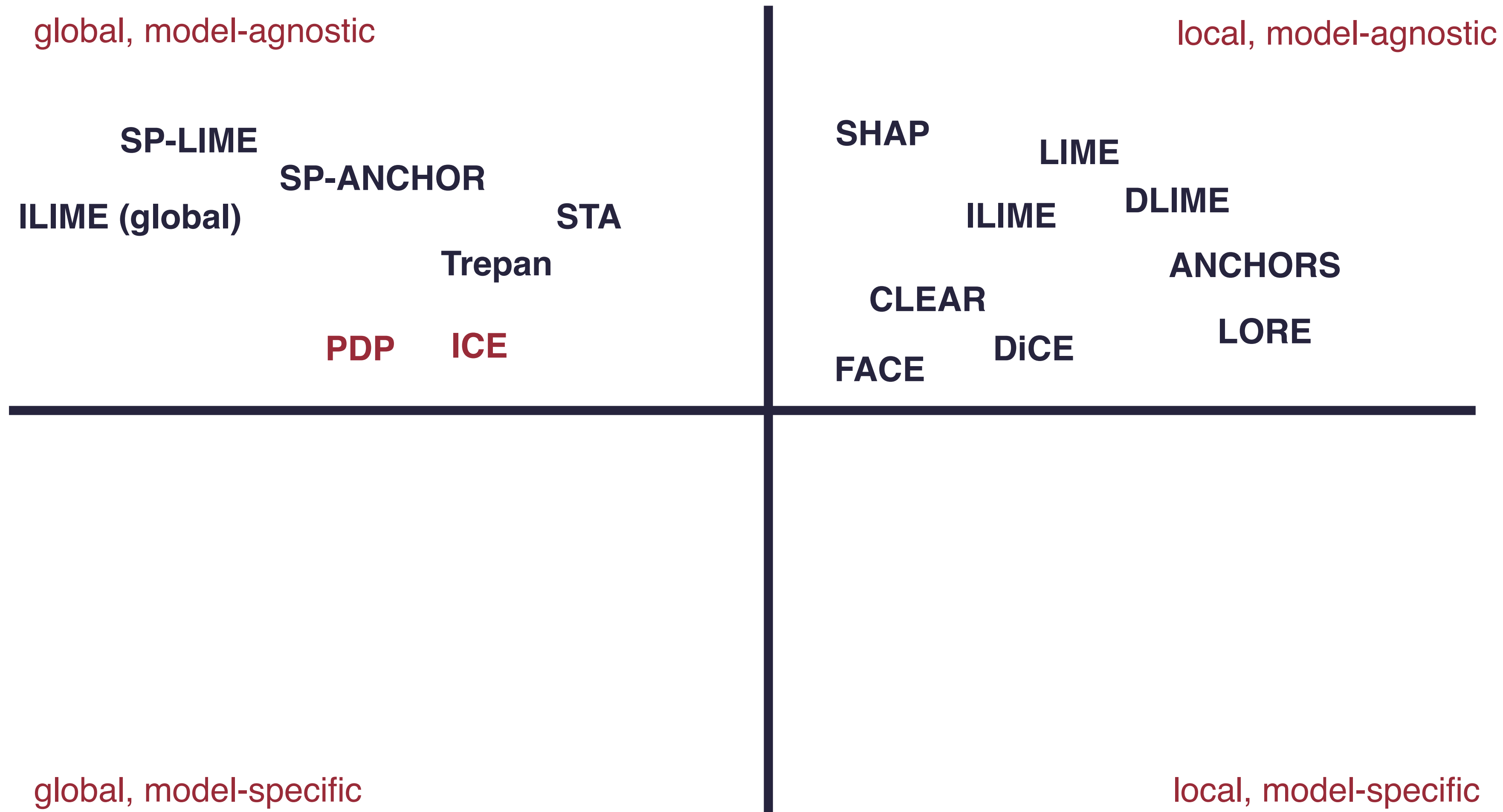
A taxonomy of XAI methods



Trepan and Single Tree Approximation

These methods build a single decision tree, of relatively small size, with the goal of globally approximating a black-box model. They differ in the way they exploit the existing model and in how they expand the tree.

A taxonomy of XAI methods



Partial Dependency Plots and Individual Conditional Expectation

These are visualization methods that allow us to highlight the influence of features on the prediction. PDP shows the marginal effect of one feature on the prediction globally. ICE does the same on each instance. One feature is changed the other are held constant. Sensitive to correlations.

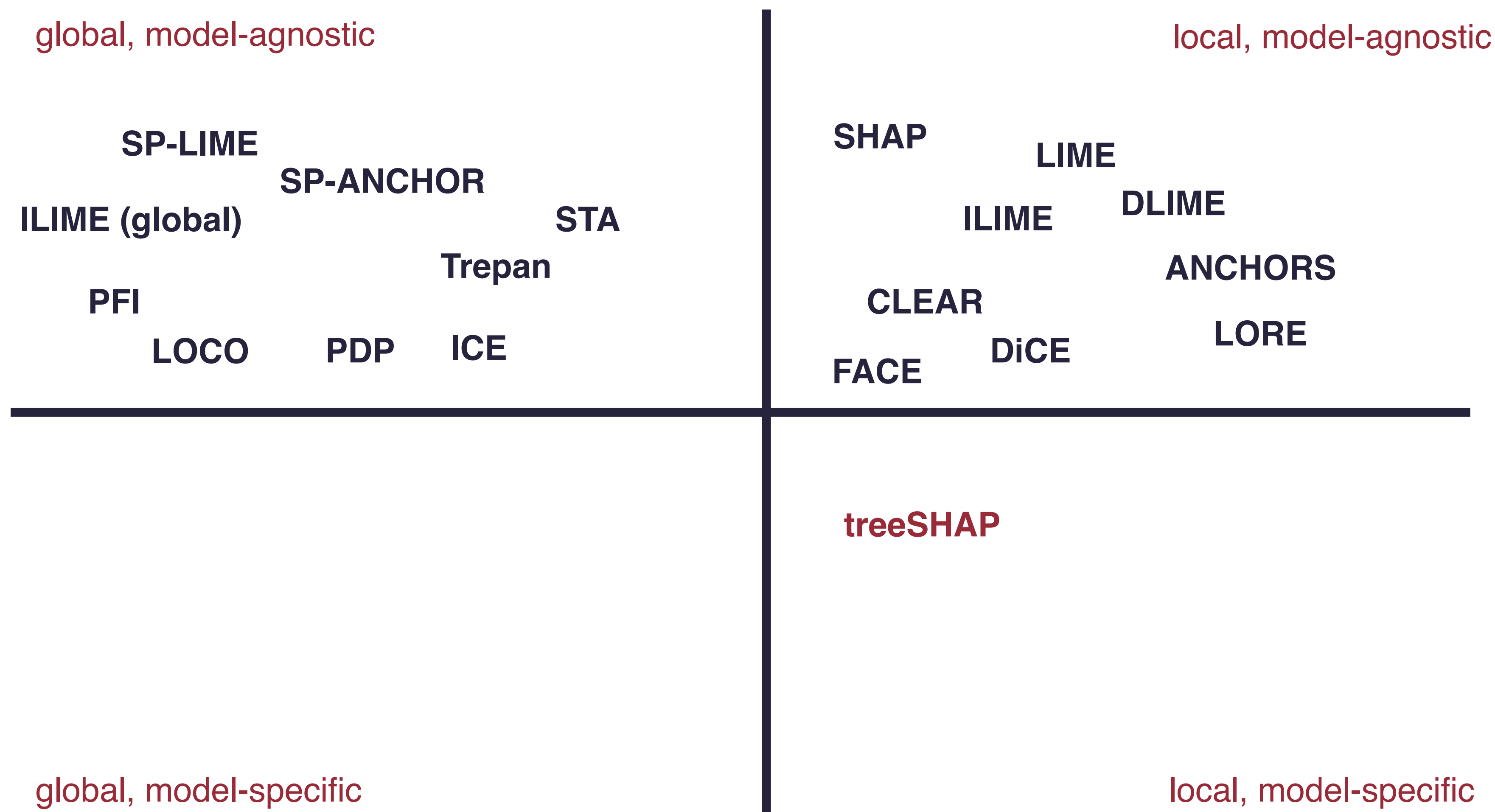
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Permutation Feature Importance and Leave-One-Covariate-Out

These methods assess the importance of a feature over the dataset by permuting or removing a feature and comparing the increase in prediction error. Sensitive to correlation among features.

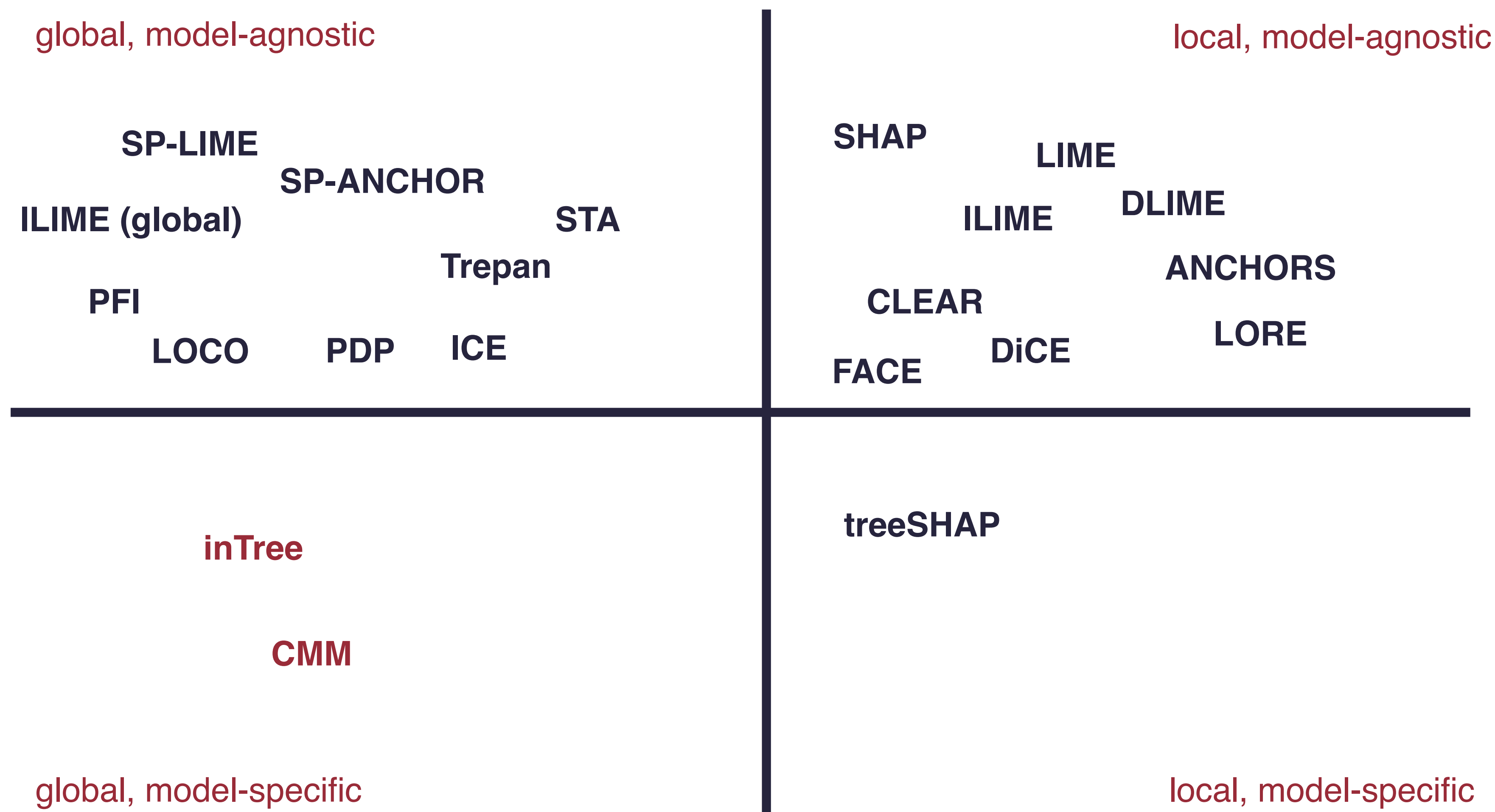
A taxonomy of XAI methods



tree SHapley Additive
exPlanation

A version of SHAP tailored to
tree based models, for which
the computation of SHAP
values is in P.

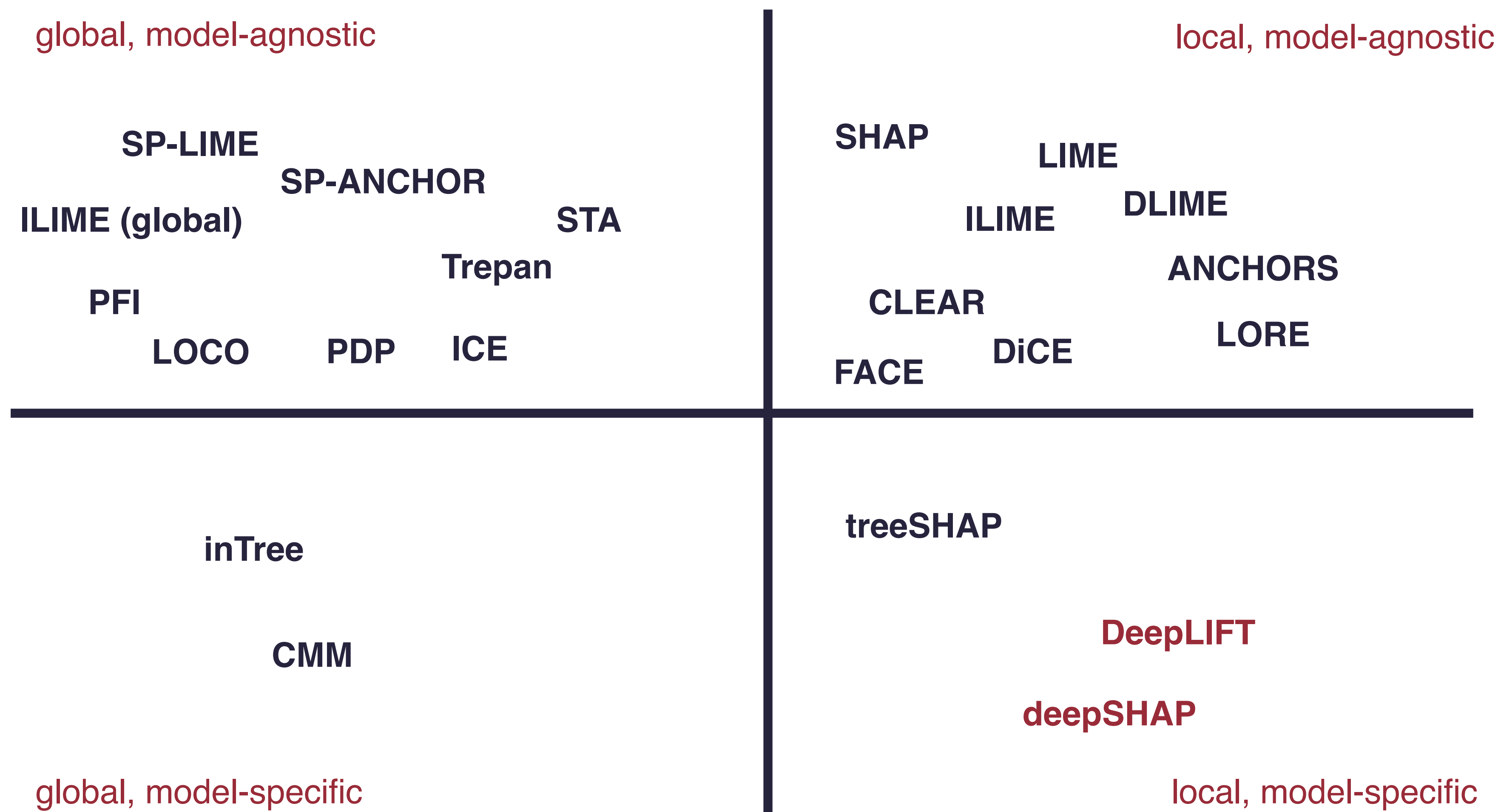
A taxonomy of XAI methods



Combined Multiple Models and interpretable Trees

These are methods designed to summarise a random forest predictor by a single tree / rule set, controlling complexity and accuracy of the approximation.

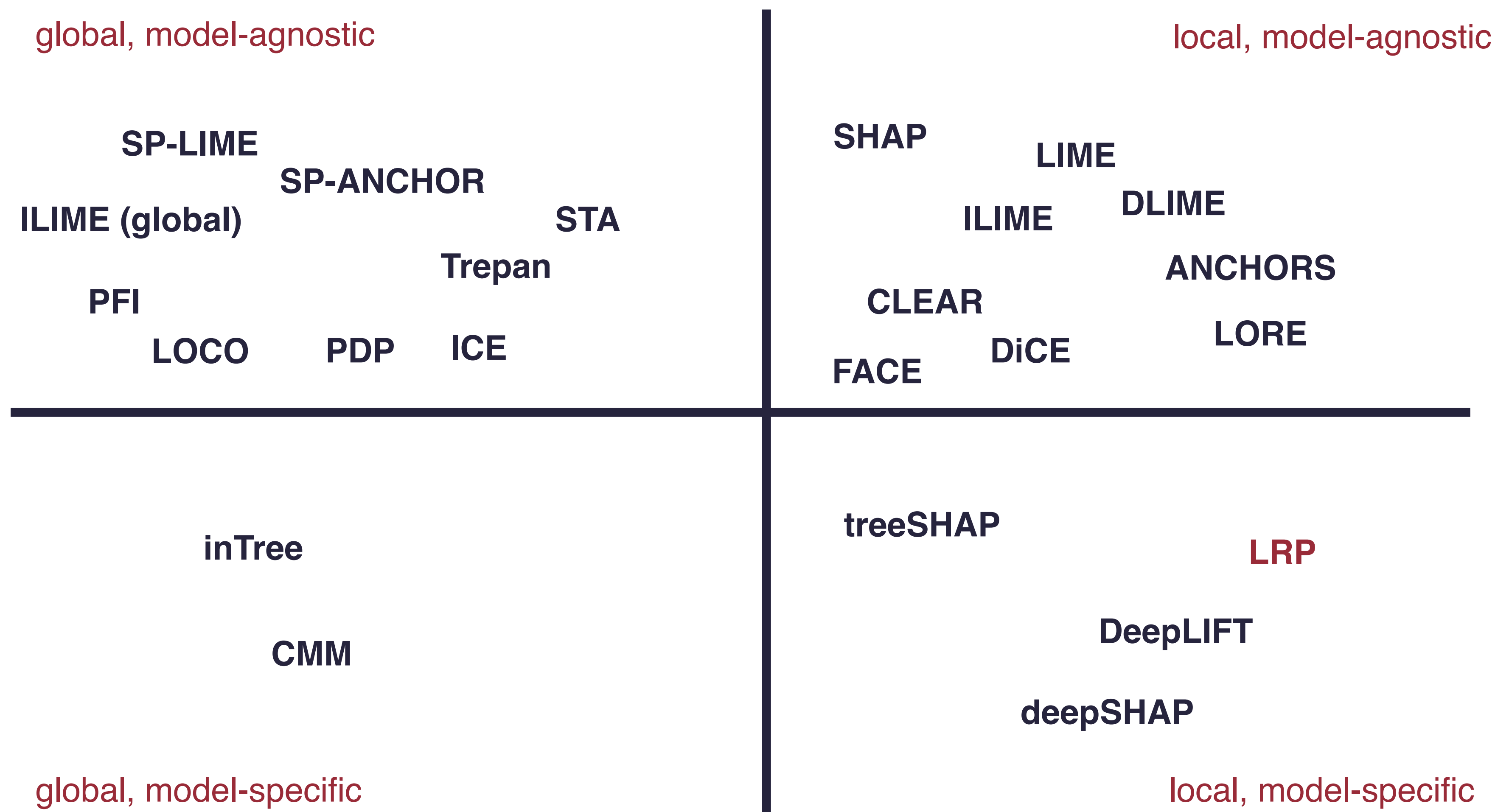
A taxonomy of XAI methods



Deep Learning Important
FeaTures and deepSHAP

deepSHAP allows to compute more efficiently SHAP coefficients for neural networks. DeepLIFT instead computes the importance of each feature as the contribution given by the feature in the difference between the predicted input and a reference input (hence it is related to SHAP).

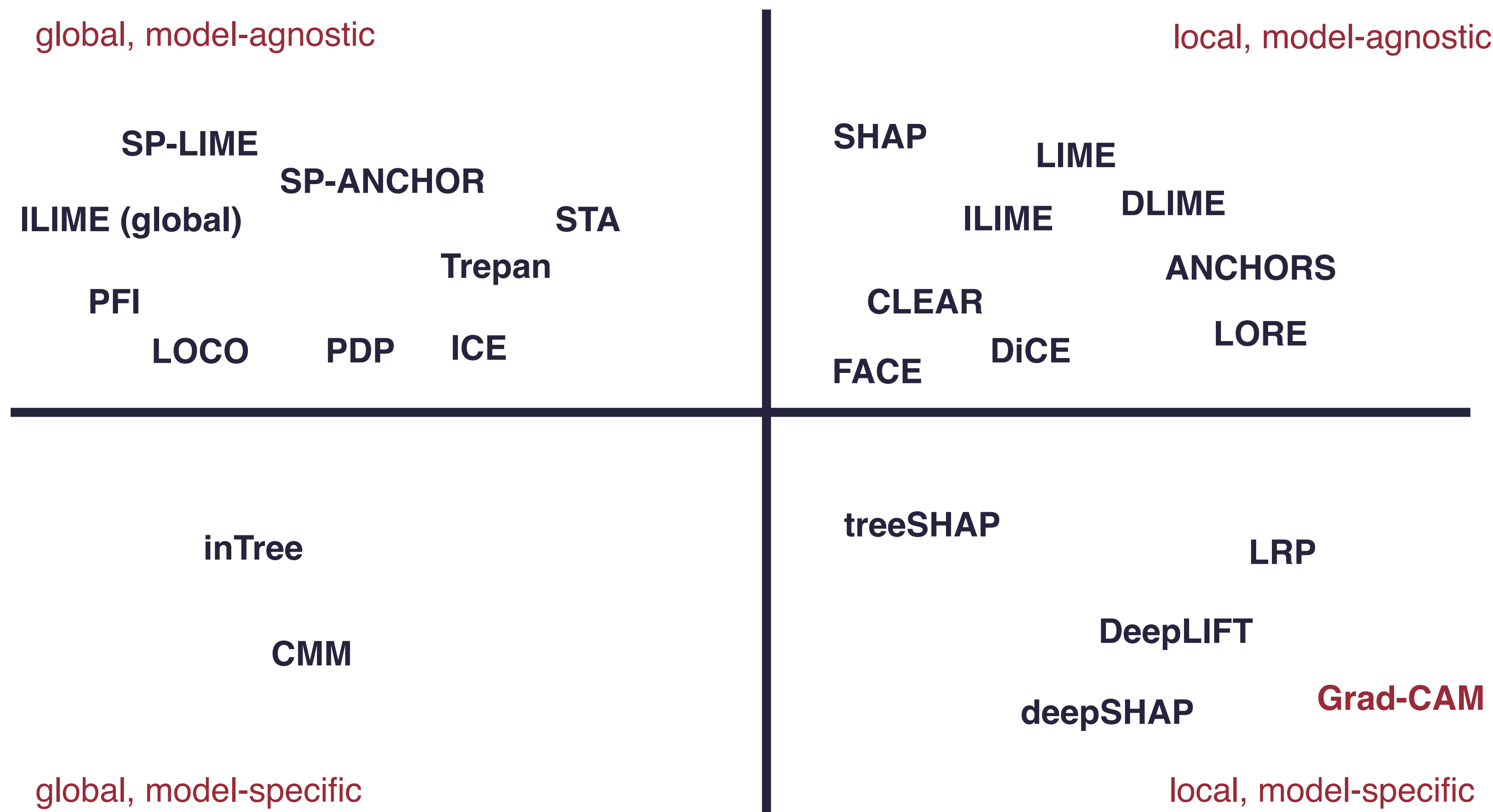
A taxonomy of XAI methods



Layerwise Relevance Propagation

Computes the importance of each feature for the prediction of an input by back-propagating the contribution of each neuron in the prediction of a class or a value proportionally to the activations of each neuron. This is also a way to decompose the prediction among the input features.

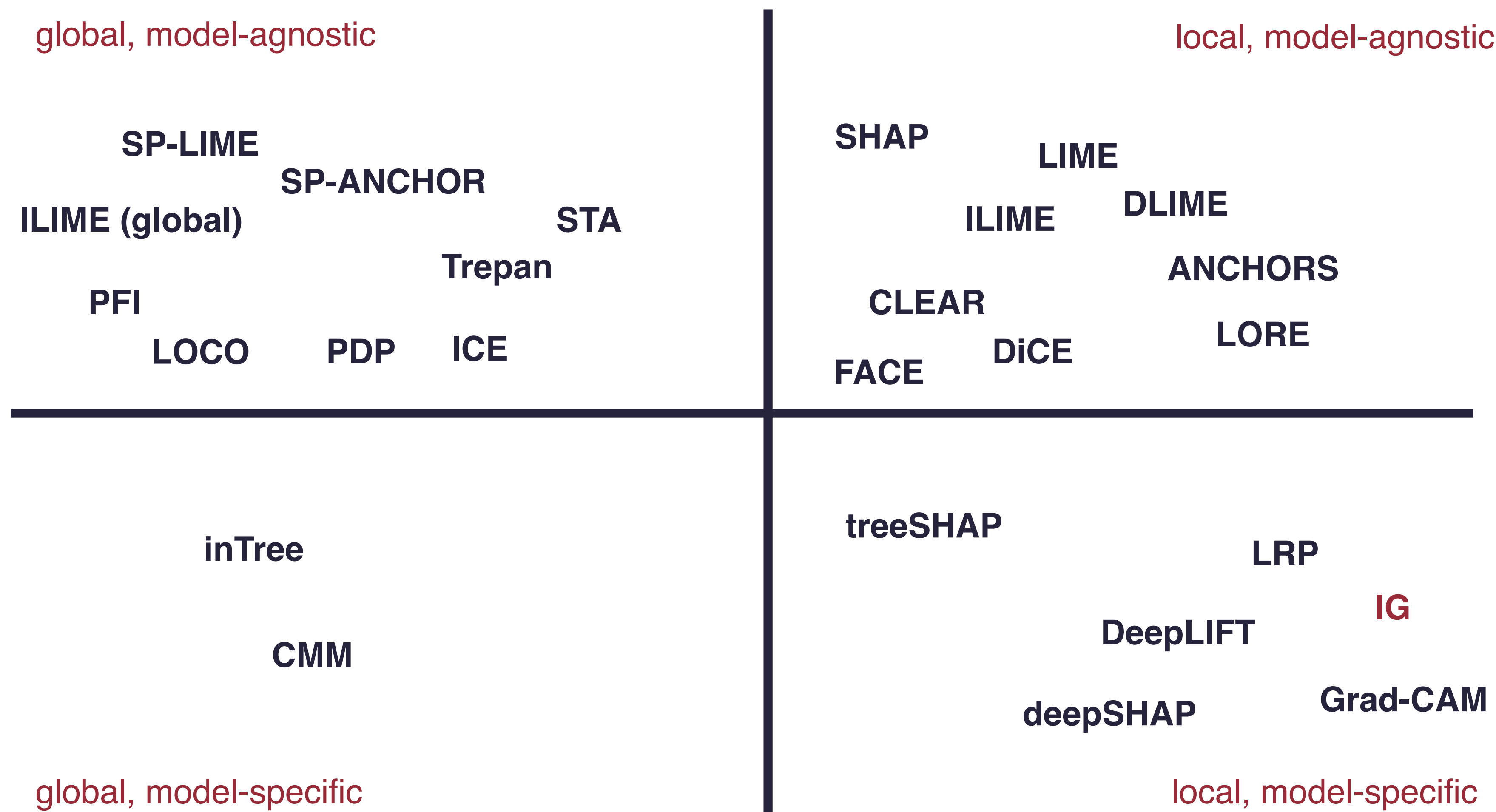
A taxonomy of XAI methods



Class Activation Maps

Grad-CAM uses the gradient information of the last convolutional layer of the CNN (gradient of the output of a chosen class w.r.t. activations), and propagates it backwards to assign importance values to each neuron (input feature) for the prediction of the considered class.

A taxonomy of XAI methods



Integrated Gradients

Works similarly to LRP but considers a sequence of images/ data points obtained by interpolation from a baseline to the input of interest. The gradient-based scores are then integrated between these different images.

Families of XAI methods (1/2)

Family	Main question	Example methods
Feature attribution and importance	Which input features matter for a prediction or model performance?	SHAP, Integrated Gradients, Grad-CAM, permutation importance
Feature effects and interactions	How do predictions vary with features, individually and jointly?	PDP, ICE, Shapley–Taylor interactions
Training-data attribution and valuation	Which training examples influence a prediction or model performance?	Influence functions, Data Shapley, datamodels, TRAK
Example-based explanations	Which representative or contrasting cases explain the data or predictions?	MMD-critic, ProtoPNet
Counterfactuals and contrastive explanations	Why this outcome rather than another? What changes could alter it?	DiCE, FACE, CLEAR, causal recourse

Families of XAI methods (2/2)

Family	Main question	Example methods
Concept-based explanations	Which human-understandable concepts are relevant to the prediction?	TCAV, ACE, concept bottleneck models
Interpretable surrogates and rule extraction	What simple model or rules approximate the decision behaviour?	LIME, Anchors, TREPAN, inTrees
Formal and symbolic explanations	Which conditions guarantee an outcome, or which symbolic steps derive it?	Abductive explanations (SAT/SMT), proof traces, neuro-symbolic reasoning
Internal representations and mechanisms	What information is encoded internally? Which computations produce the output?	Probing, feature visualisation, activation patching, circuit discovery

Course structure (XAI)

- 11 lectures + introduction
- Almost each lecture has a lab, for individual experimentation
- Bring laptops every time, in any case
- Typically XAI on Monday and Friday, until it is over. May change on short notice.
- Neurosymbolic AI is a partially independent block and will start later

NEXT LECTURES

1. Introduction
2. PDP, ICE, metrics
3. Lime, Anchors
4. SHAP
5. Rule-based methods
6. Counterfactuals
7. Gradient-based methods, focus on tabular data
8. GradCAM and extensions, focus images
9. Concept-based methods
10. Explainable methods for transformers
11. Robustness of explanations and adversarial attacks
12. Neuro-symbolic explainability (may come later in the semester)

EXAM

- Group project (2-3 people).
- one project for XAI, one for neurosymbolic, or a joint, larger project.
- Topic of our choice, investigating a method not seen in the course, or critical comparison.
- Oral presentation, 10 minuted max, with slides, plus discussion.
- You should be able to reason and **answer** simple **questions** about topics of lectures.

THIS IS THE END



Thanks for the attention!